picBunch v0.9.1: a free-space FFT convolution for the initial field of a uniformly moving particle bunch, and its measured cost and accuracy

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1 Introduction

When a particle-in-cell (PIC) simulation starts, the bunch of particles must carry an initial electromagnetic field, or the first push already acts on the wrong force. The field of a moving bunch in free space is known in closed form for each point charge, but a simulation reads that field on a staggered mesh with thousands of observation points, so it is computed by a fast sum over the source charges rather than by evaluating the closed form everywhere. We compute it as a free-space FFT convolution of a TSC deposit with the sampled TSC-convolved point-charge kernel. The field is the present-time field of uniformly moving charges, with no radiation term and no history. This paper gives the exact discrete recipe a re-implementer needs, and it measures the cost and the accuracy of `picBunch v0.9.1` (commit `c0bf82c`, tag `262927ba`) on a V100, an H100 and a gfx1100, naming the reference for every error statement. The release is driven by a single TOML configuration (input file, destination, config; Section 3.1), and it carries the capabilities the measurements below exercise: an opt-in near shell with a fast closed-form evaluator, MPI component-parallel mesh and tile-parallel near shell with bitwise equality across rank counts, a tolerance-derived far-tail margin for zero fill, and a float32 checkpoint that restarts under a stock PIConGPU.

The established workaround avoids computing the initial field altogether: one starts from a field-free state and marches the bunch forward, letting the fields build up in the first few steps. The present-time field of a uniformly moving bunch has no transient, so this history has no physical basis. We measure how well the discrete method reproduces that field and how much it costs, and we state the conditions under which the measured error is largest (Section 5).

2 Field definition

The computed quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p, and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation point $\mathbf{x}_{\mathrm{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\mathrm{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\mathrm{point},c}\big(\mathbf{x}_{\mathrm{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 3) and K_{point} is the per-unit-charge field of one uniformly moving charge. Because the integral is linear in the source charge, the sum over particles is exact; the whole design is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge [extraction worker, 2026]. For constant velocity the field points to the instantaneous position. In the lab frame, with $\mathbf{R} = \mathbf{x}_{\mathrm{obs}} - \mathbf{x}_{\mathrm{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \mathrm{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2\, \mathrm{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \mathrm{invg}^2\, \mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class (Section 6.16); the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The reference normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta}\times E_n$ [extraction worker, 2026]. The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m and $c = 299792458$ m/s. The h^3 factor (Section 3) is the Riemann volume of the sampled continuum integral [extraction worker, 2026].

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis [extraction worker, 2026]:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Method

3.1 The operator interface

`v0.9.1` (commit `c0bf82c`, tag `262927ba`) is driven by `picBunch -input bunch.h5 -destination <checkpoint> -config params.toml`, with every unit and conversion parameter in the TOML file [placement worker, 2026]. The `[picongpu]` section carries `unit_length`, `unit_e`, `unit_b`, `dt`, `step`, `precision`, `species`; `[domain]` carries `cells`, `cell_size`, `origin`; `[placement]` carries `offset_unit` and the `left`/`front`/`top` fractions; `[accuracy]` carries `tolerance` and the near-shell settings, `near_shell` `none`/`exact`/`fast`/`fast-gpu` plus `near_shell_radius` and `near_shell_tile`. A missing conversion key causes a clear error at start-up. The input bunch is centred by its charge-weighted centroid and placed at the requested offset from the domain origin; a placement leaving the bunch outside the domain is rejected at start-up. `patch_count` in `[placement]` splits the species into one particle patch per output rank, with cumulative, disjoint `numParticlesOffset` ranges, for multi-rank restarts (Section 7) [placement worker, 2026, mpi worker, 2026a].

The default `precision` is uniform `float32` for every field and particle record, and such a checkpoint restarts under a stock single-precision PIConGPU; `float64` is a uniform opt-in that records a warning attribute because stock PIConGPU cannot load double records [precision worker, 2026]. The `charge` and `mass` records are written rather than omitted, and the species is a single particle patch, `patch` = `"bbox"` or `"domain"` [placement worker, 2026].

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, and a free-space FFT convolution. Let N be the bunch-box nodes per axis, $M = \mathrm{pad} \cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at ih **cell**), and **cell** $= (dx, dy, dz)$. All precision-critical work is in double [scaling worker, 2026a, extraction worker, 2026].

3.2 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h [extraction worker, 2026]. The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\mathrm{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd` [extraction worker, 2026].

3.3 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\, \mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.5). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide [worker, 2026a]. Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT [extraction worker, 2026].

3.4 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution. The vendor plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transformations [extraction worker, 2026]. The unnormalized round trip is scaled by M^3; the code divides by $h^3 M^3$ in the read-out [extraction worker, 2026].

Padding. The grid must be zero-padded to pad ≥ 2. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box [worker, 2026b]. For a single stationary particle the pad $= 2$ result agrees with the direct non-cyclic sum to 3.811×10^{-16} and the pad $= 3$ result to 1.905×10^{-16} [worker, 2026b]. At pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$) [worker, 2026b]. The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error [worker, 2026b].

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real [worker, 2026b]. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components [worker, 2026b].

The h^3 factor. The sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units); the read-out therefore applies the factor $h_x h_y h_z / (M_x M_y M_z)$ [extraction worker, 2026].

3.5 Read-out

The six field components live at the staggered Yee positions, in cell units [extraction worker, 2026]:

$$\begin{array}{llllll} E_x & (0.5, 0, 0) & E_y & (0, 0.5, 0) & E_z & (0, 0, 0.5) \\ B_x & (0, 0.5, 0.5) & B_y & (0.5, 0, 0.5) & B_z & (0.5, 0.5, 0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed [extraction worker, 2026].

3.6 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units [worker, 2026a]. At a fixed distance in cells the error is therefore independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1, 1/2, 1/4, 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4 [worker, 2026a]. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The controlling length for a beamed source is the beaming cone width d/γ: where that width is sub-cell, the error is set by the cone, not by the mesh, and refinement stops paying off (Section 6.13) [worker, 2026c].

3.7 The near shell and the fast evaluator

The default mesh is the whole field. `v0.9.1` carries an opt-in near shell (`[accuracy]` in the TOML config, Section 3.1), with a radius and a tile [near-shell worker, 2026, far-tail worker, 2026, fast-near worker, 2026]. A distributed-FFT option (Section 6.10) makes the output field slab-local. It computes

$$F = F_{\mathrm{mesh}}(\mathrm{all}) - F_{\mathrm{mesh}}(\mathrm{near}) + F_{\mathrm{near}}(\mathrm{near}), \tag{9}$$

where the subtraction uses a per-tile local zero-padded FFT of the near sources (padding 2) and the near term is evaluated per (source, node, component) by one of two backends selected by `near_shell`: `exact`, a vendored adaptive FP64 TSC-convolution quadrature [near-shell worker, 2026], or `fast`, the closed-form evaluator below. The default remains `none`; the radius is in mesh nodes of the compute grid, and the envelope it must cover depends on h (Section 6.5). The fast backend is the accurate one: its measured accuracy and cost are in Sections 6 and 6.8 [fast-near worker, 2026]. The `exact` evaluator shares its vendored adaptive Gauss–Legendre quadrature with the independent driver reference, and that path overestimates moving oblique in-support points by a factor of ~ 10 at the closest node, so `exact` is retained for comparison rather than as a moving in-support truth (Section 6.5, [fast-near worker, 2026, pr24w53 ref, 2026]).

The fast evaluator splits the in-support integral of Eq. (1) into a singular part and a smooth remainder. The singular part is the eight-term `asinh` box of the kernel's divergence, O(1) per pair, built from the double integral

$$\begin{aligned} T(A,B,u) &= \int_0^A \int_0^B \frac{\mathrm{d}X\,\mathrm{d}Y}{\sqrt{u^2+X^2+Y^2}} \\ &= \begin{cases} A\,\mathrm{asinh}(B/A) + B\,\mathrm{asinh}(A/B), & |u| < 10^{-300}, \\ A\,\mathrm{asinh}\big(B/\sqrt{u^2+A^2}\big) + B\,\mathrm{asinh}\big(A/\sqrt{u^2+B^2}\big) & \\ \quad -u\,\mathrm{atan}\big(AB/(u\sqrt{A^2+u^2+B^2})\big), & \text{otherwise.} \end{cases} \end{aligned} \tag{10}$$

with $A, B > 0$ (each box face has 8 corners, 24 such calls per component, so the eight-term box is O(1) per pair). The form is even in u, so the $|u| \to 0$ branch must fire on $|u|$, not u (a negative u is a field point beyond a box face) [fast-near worker, 2026]. Verified against a converged midpoint trapezoid reference (-2.6348 converged; the naive trapezoid -2.6333 is 1.5×10^{-3} low, a trapezoid singularity artefact [fast-near worker, 2026]). The primitive is implemented in the fast evaluator header `include/picBunch/detail/FastNear.hpp` and the device reference `tools/closed-form/gpu_extract.cu` at picongpu `b3699df` [fast-near worker, 2026, closed-form worker, 2026]. The remainder is a plain tensor Gauss–Legendre quadrature on the subtracted integrand $(S - W^*)K$ over the TSC knot panels, each axis split at the TSC knots and at the anchor point s^*; the default order is 5, 8000 nodes per in-support pair [fast-near worker, 2026, evaluator worker, 2026]. For a moving bunch the box is γ-stretched for the axial direction, and the oblique direction uses a rest-frame octant quadrature of the singular box at order 8, 4096 nodes, plus the same remainder. Host and device use the same algebra [fast-near worker, 2026].

The assembly is exact. Single-source tests at a tile centre and a tile boundary give an assembly residual of 2×10^{-12} to 1.5×10^{-11} SI (4×10^{-16} relative), flat with distance. A local FFT of the near set against the global FFT agrees at 10^{-15} for margins 0, 2, 4 and a 100-particle cloud, and boundary accounting shows the same near set feeding the exact sum and the subtracted mesh (field identity 3.6×10^{-13}). A direct reproduction with 100 particles at radius 3 gives an assembled field exact to 4.6×10^{-13} over the whole grid, while the plain mesh on the same plane is wrong by 5.6×10^{-1} [worker, 2026d].

The fast backend changes no far number: the mesh-only field from the fast and the exact backends is bitwise identical (same md5), and the total field outside the near tiles is bitwise equal, so the tile subtraction stays intact [fast-near worker, 2026]. The seam jump excess at the fast/exact crossing, on a 16^3 grid, $R = 3$ cells, tile 4, cells 10^{-7} m, is 2.03×10^{-10} per component for a stationary single particle (2.02×10^{-10} on $E_x/E_y/E_z$, zero on B) and 2.8×10^{-10} for the 64-source cloud [fast-near worker, 2026, release worker, 2026].

3.8 Bunch and grid configurations

Every case measured in the paper is listed in Table 1 with the bunch and the grid, and the section that uses it. The entries are *recorded* values transcribed from the run artifacts: the mid-bunch slice cases from

B1 (bunch files bunches/gamma{1,10}-1k/bunch.h5, read-back bunches_verification.json and the gamma{1,10}_1k_reference.h5/json driver output) and LARGE1 (data/plan.json, RESULTS.md); the cost and accuracy cases from CFPORT and INT4; the γ landscape from ER1b and the aspect-4 rows from N2b; the multi-velocity cases from MV2 and MV1b; and the divergence case from VAR1 (the VAR1b $N = 200$ subset exists, but its independent reference is pending, so no row is included). The generation line for the mid-bunch bunches is generate_bunch.py -model gamma -gamma G -seed 42 -grid 64 64 32 -cell-size 1e-7 1e-7 1e-7 -centroid 32 32 16 -sigma 6 6 6 -weights uniform [reference-field worker, 2026]. Entries that no artifact records are written as "not recorded"; nothing here is inferred. The total charge is N times $-e$ for the electron bunches with one unit weight, where $e = 1.602176634 \times 10^{-19}$ C; the kinetic energy is $(\gamma - 1)\, m_e c^2$ with $m_e c^2 = 510.999$ keV.

Table 1: Configurations of every measured case and the section that uses it (recorded values only; "not recorded" means no artifact records the value). Sources: mid-bunch slices B1; large-domain LARGE1; cost/accuracy CFPORT,INT4; γ landscape ER1b with aspect-4 rows N2b; multi-velocity MV2,MV1b; divergence VAR1. The two mid-slice entries give the bunch generation grid; the field grid for those cases ($32 \times 32 \times 16$ @ 2×10^{-7} m) is in the second block. Kinetic energy is $(\gamma - 1)\, m_e c^2$; the multi-velocity driver cell size is the MV1b value (10^{-6} m). The VAR1b $N = 200$ subset exists but its independent reference is pending, so it is not a row here.

case	N	total charge ($-e$ each)	size (µm; cells/axis)	spread	used in
mid slices, g10	1000	1.602×10^{-16} C	bunch generation grid $64 \times 64 \times 32$ @ 10^{-7} m ($6.4 \times 6.4 \times 3.2$ µm), $\sigma = 6$ cells/axis [reference-field worker, 2026]	none ($\gamma_{\min} = \gamma_{\max} = 10$) [reference-field worker, 2026]	§6.5, §6.7, §7
mid slices, g1	1000	1.602×10^{-16} C	bunch generation grid $64 \times 64 \times 32$ @ 10^{-7} m ($6.4 \times 6.4 \times 3.2$ µm), $\sigma = 6$ cells/axis [reference-field worker, 2026]	none ($\gamma_{\min} = \gamma_{\max} = 1$) [reference-field worker, 2026]	§6.5, §6.7, §7
large-domain, g10	100	1.602×10^{-17} C	extent $2.74 \times 3.78 \times 2.65$ [large-domain worker, 2026]	not recorded	§6 (Fig. 11)
large-domain, g1	100	1.602×10^{-17} C	extent $2.74 \times 3.78 \times 2.65$ [large-domain worker, 2026]	not recorded	§6 (Fig. 13)
cost, stationary	2×10^6	not recorded	20-cell bunch [release worker, 2026]	none	§6.8
cost, moving axial	2×10^6	not recorded	20-cell bunch [release worker, 2026]	none	§6.8
landscape, aspect 1	1	1.602×10^{-19} C	single particle, 96-cell box [worker, 2026c]	none	§6.13
landscape, aspect 4	1	1.602×10^{-19} C	single particle at the origin [worker, 2026a]	none	Table 7
multi-velocity, E 5%	512	not recorded	3-cell position cloud [multi-velocity worker, 2026a]	5% energy (seed 100) [multi-velocity worker, 2026a]	§6.16 (superseded)
multi-velocity, A 5°	512	not recorded	3-cell position cloud [multi-velocity worker, 2026a]	5° angular (seed 103) [multi-velocity worker, 2026a]	§6.16 (superseded)
multi-velocity, E 5% (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$ [multi-velocity worker, 2026b]	5% energy, $+y$ drift [multi-velocity worker, 2026b]	§6.16
multi-velocity, A 5° (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$ [multi-velocity worker, 2026b]	5° angular, $+y$ drift [multi-velocity worker, 2026b]	§6.16
1 mrad, binned (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$, 1.046 mrad rms x-div [multi-velocity worker, 2026c]	1 mrad, $+y$ drift [multi-velocity worker, 2026c]	§6.16

case	N	total charge ($-e$ each)	size (µm; cells/axis)	spread	used in
divergence, 1 mrad	1000	1.602×10^{-16} C	$\sigma = 0.6$ each axis, centroid $(3.197, 3.208, 1.615)$ [divergence worker, 2026]	1 mrad paraxial, $+y$ (seed 43) [divergence worker, 2026]	§6 (reference pending)

case	γ, kinetic E ($m_e c^2 = 510.999$ keV)	weight, cell	field grid (domain), host
mid slices, both	10 / 5.06 MeV; 1 / 0	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ ($6.4 \times 6.4 \times 3.2$ µm), rosi_v100
large-domain, both	10 / 5.06 MeV; 1 / 0	not recorded; 2×10^{-7} m	$128 \times 128 \times 64$ ($25.6 \times 25.6 \times 12.8$ µm), jureca-h100
cost, both	1 / 0; 10 / 5.06 MeV	not recorded; 2.5×10^{-7} m	$66 \times 68 \times 66$, jureca-h100
landscape, aspect 1	10/50/100	1; 10^{-7} m	96^3 (h=1), 384^3 (h=1/4), jureca-h100 (ref: rosi_v100)
landscape, aspect 4	1/10/100	1; anisotropic, aspect 4 [worker, 2026a]	axis-aligned, $h = 1$, host not recorded [worker, 2026a]
multi-velocity, both	10 / 5.06 MeV	not recorded; 10^{-6} m [multi-velocity worker, 2026d]	48^3, jureca-h100
multi-velocity, both (v0.9.1)	10 / 5.06 MeV	1 each; 10^{-7} m	64^3, hal (A30, `sm_80`) [multi-velocity worker, 2026b,c]
divergence	10 / 5.06 MeV	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ ($6.4 \times 6.4 \times 3.2$ µm), rosi_v100

The mid-bunch bunches are mono-energetic: the read-back records $\gamma_{\min} = \gamma_{\max} = 1$ for the $\gamma = 1$ bunch and $\gamma_{\min} = \gamma_{\max} = 10$ for the $\gamma = 10$ bunch [reference-field worker, 2026]. The MV2 multi-velocity cases (48^3, 512 sources, `02891fe6`) were superseded by the v0.9.1 re-measurement on a 64^3 grid with 200 sources (`BDATA2` bunches, 10^{-7} m cell, `hal` A30) [multi-velocity worker, 2026b,c]; the v0.9.1 numbers in Section 6.16 replace the MV2 binning-error table. The MV2 campaign bunch files were unavailable; the per-particle sample was regenerated from the surviving driver block at `02891fe6` (seeds 100, 103) on `rosi_v100`; the LCG state and draw count are identical to the campaign, and only last-bit `libm` differences between hosts are possible, so the sampled distributions are statistically identical.

3.9 MPI parallelism

`v0.9.1` writes a global output domain larger than the compute box [mpi worker, 2026b, far-tail worker, 2026], and computes the field in parallel across ranks [rank-parallel worker, 2026]. The written extent is set by `-global-grid` and `-global-origin`. The global array is never allocated: each rank owns a contiguous z-slab split into tiles of about 128 MiB, zero-filled, then filled with the compute-box overlap, the rest exact 0.0. All ranks open one openPMD series with `MPI_COMM_WORLD` and write disjoint `storeChunk` regions with collective flushes, so the output is a single dataset, not per-rank files. A `-deterministic-deposit` flag gives a serial host deposit for bitwise comparisons [mpi worker, 2026b].

The compute is parallel in two ways, neither of which is a domain-decomposed FFT [rank-parallel worker, 2026]. The mesh runs six independent component convolutions: every rank deposits and transforms the same ρ spectrum (replicated), rank r evaluates the components c with $c \bmod \text{size} = r$, and one `MPI_Allreduce` sums the six arrays. The near shell partitions the output tiles by linear tile index; each rank builds its own near-source list and local subtraction FFT for its tiles and writes only those corrections, and one `MPI_Allreduce` sums the correction arrays. The bunch is read once on rank 0 and broadcast, not redistributed. Because the mesh components are disjoint and each near correction element is owned by exactly one rank, the all-reduce order cannot change a value: the outputs are bitwise identical at 1, 2 and 4 ranks (Section 4.1) [rank-parallel worker, 2026, mpi worker, 2026a].

3.10 The far-tail margin

For a global domain that extends beyond the compute box, the zero fill must not start until the field is below tolerance. `v0.9.1` places that boundary at the smallest centre distance d where

$$E_{\text{tail}}(d) = \gamma_{\max} k \left(\frac{|Q|}{d^2} + \frac{2|\mathbf{p}|}{d^3} + \frac{3.5\,\|D\|_2}{d^4} \right) \le \text{tolerance} \cdot E_{\text{peak}}, \tag{11}$$

with Q the total charge, $\mathbf{p}$ the dipole, $D_{ij} = \sum q\,(3x_i x_j - |\mathbf{x}|^2 \delta_{ij})$ the traceless quadrupole from the actual particle positions, $\|D\|_2$ its largest absolute eigenvalue, and $\gamma_{\max} = \max \sqrt{1 + |\mathbf{u}|^2}$ [far-tail worker, 2026]. Beyond a measured switch radius the writer instead evaluates the multipole field directly (`FarTail.hpp`), so the shipped file has no interior zero fill [far-tail worker, 2026]. The margin is reported at run time as a bound in V/m.

For the documented 20-cell, 4096-electron bunch (source radius $R = 16.85$ cells, peak 45722.5 V/m, tolerance level 45.72 V/m), the quadrupole margin is 350 cells, which exceeds the 2^{28} padded-node cap, so the code falls back to the measured far-tail switch. Table 3 gives the switch error: the worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum in 13 directions [far-tail worker, 2026].

A shipped 512^3 output sampled outside the compute box confirms the switch in practice: 36 nodes including corners, worst $|\mathbf{E}_{\text{file}} - \mathbf{E}_{\text{direct}}| = 2.8 \times 10^{-3}$ V/m against the 45.7 V/m tolerance level (6.1×10^{-5} of it), and every sample nonzero, i.e. no interior zero fill [far-tail worker, 2026].

Table 3: Far-tail switch validation, 20-cell 4096-electron bunch, $R = 16.85$ cells, tolerance level 45.72 V/m (10^{-3} of the peak 45722.5 V/m). Worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum over 13 directions [far-tail worker, 2026]. The switch at $6R$ is the one shipped: the error is 1.87×10^{-3} of the tolerance level.

switch radius	worst err (V/m)	/ tolerance level
$2R$ (33.7 cells)	4.80×10^{1}	1.05
$4R$ (67.4 cells)	8.77×10^{-1}	1.92×10^{-2}
$6R$ (101.1 cells)	8.53×10^{-2}	1.87×10^{-3}
$8R$ (134.8 cells)	1.66×10^{-2}	3.63×10^{-4}

Table 4: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0 [scaling worker, 2026a]. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

4 Measured scaling

All timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0 [scaling worker, 2026a]. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double [scaling worker, 2026a]. Each configuration ran three timed repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max [scaling worker, 2026a]. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run [scaling worker, 2026a]. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 5, a different measurement path.

Reproducibility of Table 4. The matrix is the `picBunchBench` sweep of picBunch branch experimental/bench at 3d05752a493e12f3a0a03c6754f92f202a70a482 (picBunchBench--sweep--repeat3--csvmatrix.csv); the bench source and `docs/benchmarking.md` are on that branch [scaling worker, 2026a]. The input bunches are not checked into the repository; the bench source at `3d05752a` is the generator [scaling worker, 2026a]. The fast GPU kernel tools/closed-form/gpu_extract.cu lives in picongpu at `b3699df` [fast-near worker, 2026]. The H100 runs used CUDA 13.4 [fast-near worker, 2026, release worker, 2026].

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold [scaling worker, 2026a,b]. This confirms the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff.

Fig. 1 sweeps one axis at a time: the top panel fixes the 20-cell box and varies N, the bottom panel fixes $N = 2{\cdot}10^6$ and varies the box. The scaling is in two regimes, with the crossover set by the box, not the particle count [scaling worker, 2026a]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms) [scaling worker, 2026a]. At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× because the deposit stays constant [scaling worker, 2026a]. The per-h totals are 64.76, 69.27, and 151.51 ms for $h = 1$, 1/2, 1/4 (padded array $M = 60/100/180$, 2×10^6 particles, 20-cell box) [scaling worker, 2026a]. On the `H100` (`jureca-h100`, `sm_90`), the same 2×10^6, 20-cell, $h = 1$ mesh is 0.052 s compute (deposit 34.4 ms, kernel 1.08 ms, FFT 7.44 ms, read-out 9.02 ms; medians of three), and 0.755 s CLI wall including the particle read and the HDF5 write [release worker, 2026].

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB [scaling worker, 2026a]. Memory scales with the padded bunch box, not the global field.

Table 5 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 → 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 4; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer [worker, 2026e, scaling worker, 2026a].

Table 5: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, shipped CLI `-tolerance` 10^{-3} [worker, 2026e]. A different measurement path from the compute-only Table 4: the totals include the writer (the dominant added term); the per-stage split is deposit/kernel/FFT/readout/writer. The $N = 10^6$ total is 47.15 ms here versus 61.92 ms in the compute-only campaign.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

Table 6: Rank scaling on `rosi_v100`, idle window, medians of three [mpi worker, 2026a]. Mesh: $N = 2\times10^6$, 112^3, component-parallel (replicated deposit and source FFT). Near shell: $N = 128$, 32^3, $R = 1.5$, tile 8, tile-parallel. Write: $N = 2 \times 10^6$, 256^3 global, 0.77 GB, collective HDF5. Mesh and near shell are bitwise identical at 1/2/4 ranks.

	near shell			mesh		write	
ranks	wall (s)	speedup	eff.	wall (ms)	speedup	wall (s)	speedup
1	36.61	1.00	100%	522.2	1.00	7.41	1.00
2	29.60	1.24	62%	436.3	1.20	4.34	1.71
4	27.29	1.34	34%	433.6	1.20	15.49	0.48

4.1 MPI rank scaling

The rank-parallel compute (Section 3.9) and the chunked writer were measured on an idle `rosi_v100` window, medians of three runs [mpi worker, 2026a]. The node was checked idle before the block (no CPU worker, 1-min load below 2), but each rank block starts from the preceding measurement, so the 1-min load rises within the block: the near-shell block starts at 1.9, 8.4 and 11.8 for 1, 2 and 4 ranks (mesh block 1.4–1.5, write block 1.1–1.6); no foreign process was active during the window [mpi worker, 2026a]. Correctness was verified at 1, 2 and 4 ranks on a 128-particle, 32^3, cell 8×10^{-7} m, tolerance 10^{-3}, near-shell $R = 1.5$, tile 8 problem: the mesh and the near shell are bitwise identical at 1-vs-2 and 1-vs-4, all six components, $\max_{\text{abs diff}} = 0.0$ [mpi worker, 2026a, rank-parallel worker, 2026].

Fig. 2 shows the speedup for the three parallel pieces. The mesh is component-parallel: the replicated deposit (307–332 ms, flat in rank count) and the replicated source-spectrum FFT cap it, so the speedup plateaus at 1.20× at both 2 and 4 ranks. The near shell is tile-parallel: the static round-robin tile split plus the all-reduce of the full correction array leave static-tile imbalance that plateaus it at 1.34× at 4 ranks. The writer is a collective HDF5 write: it improves to 1.71× at 2 ranks and falls to 0.48× at 4 ranks, where the collective flush and metadata traffic contend on the shared filesystem, an I/O limit rather than a compute limit [mpi worker, 2026a].

5 Measured error structure

The measured error depends on the observation point, the direction relative to the beam, and γ. We report the regimes in which the error is largest, together with the configuration and reference for each number.

The fast near backend, against the independent continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, and 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior, 1.11×10^{-4} face, 2.12×10^{-6} outside; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend at its default order (Section 6.6) [fast-near worker, 2026, release worker, 2026]. The oblique value is the one region where the fast backend is above 10^{-3} at its default order. Against the order-12/refine-3 continuum arbiter the fast backend is 1.6% at the closest moving in-support node, the arbiter-validated in-support accuracy, whereas the driver and the `exact` backend overestimate that node by $\sim 10\times$ (Section 6.5) [pr24w53 ref, 2026].

The mesh alone, without the near shell, reproduces the field to 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the slices cut the bunch core (Section 6.5); these planes cut the sources, the region where the single-mesh spectral convolution has no near-field treatment. With the near shell enabled at the measured radii ($R = 3$ for $\gamma = 1$, $R = 4$ for $\gamma = 10$ in mesh nodes of the $h = 1$ slice grid), the assembly residual is at roundoff and the measured plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$ (Section 6.7); the residual is the mesh contribution outside the shell radius.

In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M-T$ error at $d = 48$ is 2.6×10^{-4}, the $M-P$ error is 1.5×10^{-2}, and the TSC smoothing $T-P$ is 1.48×10^{-2}, of which the $M-P$ error is the TSC smoothing, not the deposit (Table 7) [worker, 2026a, Orchestrator, 2026a,b]. At $\gamma = 100$, aspect 1, the perpendicular $M-T$ error at $d = 48$ is 3.75×10^{-1} and does not decrease with further increase in d [worker, 2026a, Orchestrator, 2026c]. At aspect 4, $\gamma = 1$, the perpendicular $M-T$ error at $d = 48$ is 2.4×10^{-3} [worker, 2026a]. The axial direction is below 3×10^{-4} at $d = 6$ for every γ [worker, 2026a].

The γ-landscape measurement at $d = 8$ cells, $h = 1$, shows the error 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$. The h-refinement exponent p (error $\sim h^p$, fitted from $h = 1$ and $h = 1/4$) falls from 1.85 to about 1.0 as γ rises; the controlling length is the beaming cone width d/γ, and at $\gamma \geq 50$ it is sub-cell (Section 6.13) [worker, 2026c]. The detailed discussion

Table 7: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; single-component worst: $\gamma = 1$, aspect 4, $M-T$ 2.65×10^{-5}; $\gamma = 10$, aspect 1, 1.4×10^{-3}) [worker, 2026a]. $M-T$: mesh vs TSC-convolved continuum; $M-P$: mesh vs point field; $T-P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1 the $M-P$ vs $M-T$ gap is the smoothing (L2 ratio 58×), not the mesh.

γ	aspect	$M-T$	$M-P$	$T-P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

is in Section 8.

6 Accuracy

Each error figure names its reference (M mesh, T continuum, P point). The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

6.1 The reference and the metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map [pr24w53 ref, 2026]. Its quadrature error is below the 3×10^{-4} gate in every configuration measured here [worker, 2026a].

Three metrics are reported and are not interchangeable, and each claim in this section names which one supports it [worker, 2026f].

Floored. $|\text{ref} - \text{mesh}| / \max(|\text{ref}|, |\text{mesh}|)$, over points where $|\text{ref}| \geq 10^{-3}$ of the local (six-component) field peak. This is the relative per-component error of record for the electric field.

Unfloored. The same ratio at every point. It is the only one that reaches the magnetic field, because the floor excludes it (below).

Absolute. $|\text{ref} - \text{mesh}| / \max_{\text{point}} |\text{ref}_c|$, the peak-normalized error; it supports whole-field statements.

The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, \text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the $+1$ denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate [worker, 2026e].

The floor hides the magnetic field. For $\gamma = 10$ every B point is excluded by the floor: $|\mathbf{B}|$ is about 10^{-10} of the E-dominated local field, far below the 10^{-3} threshold, so the floored metric reports no magnetic accuracy at all. The unfloored B relative error is 9.7×10^{-4} at $d = 32$ (single particle, aspect 1), against a B peak that the floored column never sees [worker, 2026f]. The floored column of Table 8 is therefore electric-only at $\gamma = 10$.

6.2 Reference-separated error at 48 cells

Table 7 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M-P = 1.5 \times 10^{-2}$) is about 58× the mesh against the TSC-convolved continuum ($M-T = 2.6 \times 10^{-4}$, an L2 ratio); the difference is the TSC smoothing, which $M-P$ inherits and $M-T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M-T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ [worker, 2026a].

6.3 The independent validation

A second, independent pass measured the shipped CLI end to end: `picBunch` at `bfaa1e26`, rosi_v100, the order-12 `indfield` reference (refine 3), $d = 2$ to 32 cells, $\gamma \in \{1, 10\}$, aspect 1 and 4, single particle and three-particle cloud [worker, 2026e].

Table 8 confirms the reference-separated picture independently. The measured value is 1.1×10^{-4} at $d = 12$ for $\gamma = 1$, aspect 1, and 2.6×10^{-4} at $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1 the relative error is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$. These are floored electric values; the unfloored magnetic error at $\gamma = 10$ is 9.7×10^{-4} at $d = 32$. The window starts at $d = 2$, so the TSC-support interior ($|s| < 1.5$ cells) is not covered by these sweeps; it is what the near shell measures (Sections 3.7, 6.8) [worker, 2026e,f,d].

Fig. 3 plots the mesh against the TSC-convolved continuum; the three-particle cloud series is in Table 8. The series is floored, so the magnetic field, which the floor excludes at $\gamma = 10$ (Section 6), is not in the figure. The $\gamma = 10$ point at $d = 48$ (1.8×10^{-3}) is in Table 8 but outside the plotted window. The point-field column of Table 7 ($M-P$) is the mesh against the un-smoothed field, which carries the TSC smoothing in addition to the deposit error and never agrees with $M-T$ for $\gamma h \gg 1$.

Table 8: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the shipped CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; rosi_v100, the order-12 `indfield` reference (refine 3) [worker, 2026e]. "s/c" = single / cloud. Measured values at $d = 48$: $\gamma = 10$, 1.8×10^{-3} ($d = 32$: 6.5×10^{-3}); $\gamma = 1$ is below 3×10^{-4} from $d = 12$ (aspect 1) and $d = 32$ (aspect 4).

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

Table 9: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: beyond the measured window) [worker, 2026a]. T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger. The $\gamma = 1$ aspect 1 row is $d = 8$ against both references here, versus the $d = 12$ of the shipped-CLI series in Table 8: the two use different component sets (six axis directions, L2-pooled, versus per-component over x, y) and different implementations [worker, 2026a,e].

			3×10^{-4}		10^{-3}	
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

6.4 Crossover radii

At $\gamma = 1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d = 8$; the two are interchangeable there. From $\gamma = 10$ up, against the point field the measured perpendicular error at $d = 48$ is 6.5×10^{-3} ($\gamma = 10$, aspect 1); against the TSC-convolved continuum the perpendicular crossover is $d = 48$ ($\gamma = 10$, aspect 1), and for the remaining configurations the measured value at $d = 48$ exceeds the gate. The axial direction is 6 against the continuum at every γ [worker, 2026a].

6.5 Mid-bunch slices: field and error side by side

The distance sweeps above sample the far field. To measure the mesh in the bunch core we slice through it and compare, pixel by pixel, with an exact TSC-convolved reference: 64×64 planes, cell 10^{-7} m, $N = 200$, $Q = -100$ pC, $\gamma = 10$ with momentum along $+y$ (`BDATA2`). Two cases share the positions and differ only in a 1.045 mrad rms transverse divergence in x (Figs. 4, 5, 6, 7); a separate 32×32 plane carries the binned 1 mrad case (`VAR1b`, Fig. 8). In every panel y is horizontal and increases left to right, so the bunch moves rightwards; the vertical axis is x in the x-y cut and z in the z-y cut. Each figure has a field row with four columns (E_x, E_y, E_z and the magnitude $\sqrt{\sum_i E_i^2}$) and an error row underneath; the magnitude is the square root of $E_x^2 + E_y^2 + E_z^2$ so that it shares the components' linear scale. A row has one colour scale and one colour bar at its right edge, the error row is logarithmic and masked where the reference is below 10^{-3} of its own peak. The forward case compares the shipped mesh plus fast shell against the order-4 adaptive Gauss–Legendre driver [beam-direction worker, 2026]; that driver overestimates moving oblique in-support points by about 10×, so the solid white outline marks the TSC support and the in-support error pixels are shown against a reference that is itself unreliable there. The trustworthy in-support statement is the fast-versus-arbiter 1.6% at the worst node (Section 6.6). The fluctuation reference is complete, but picBunch v0.9.0 rejects a momentum spread, so only its reference field panels are shown and the picBunch panels are labelled unavailable.

Shell radius unit. The earlier $N = 1000$ slices run the default mesh with no near shell. The near-shell envelope measured in the crossover work, "3 to 4 PIC cells," assumed a mesh spacing $h = 1/4$ of the PIC cell, so $R = 3$ and $R = 4$ are 12 and 16 mesh nodes [near-shell worker, 2026, slice worker, 2026a]. On this grid the node spacing is $h = 1$ PIC cell (2×10^{-7} m), so the same $R = 3/4$ is 3/4 mesh nodes, four times smaller in physical units than the envelope configuration.

At $R = 3$ on an $h = 1$ grid the shell covers 3 mesh nodes; the TSC support extends ± 1.5 cells from each source, so the shell covers a fraction of the full support. The measured plane errors at this radius are reported in Section 6.7.

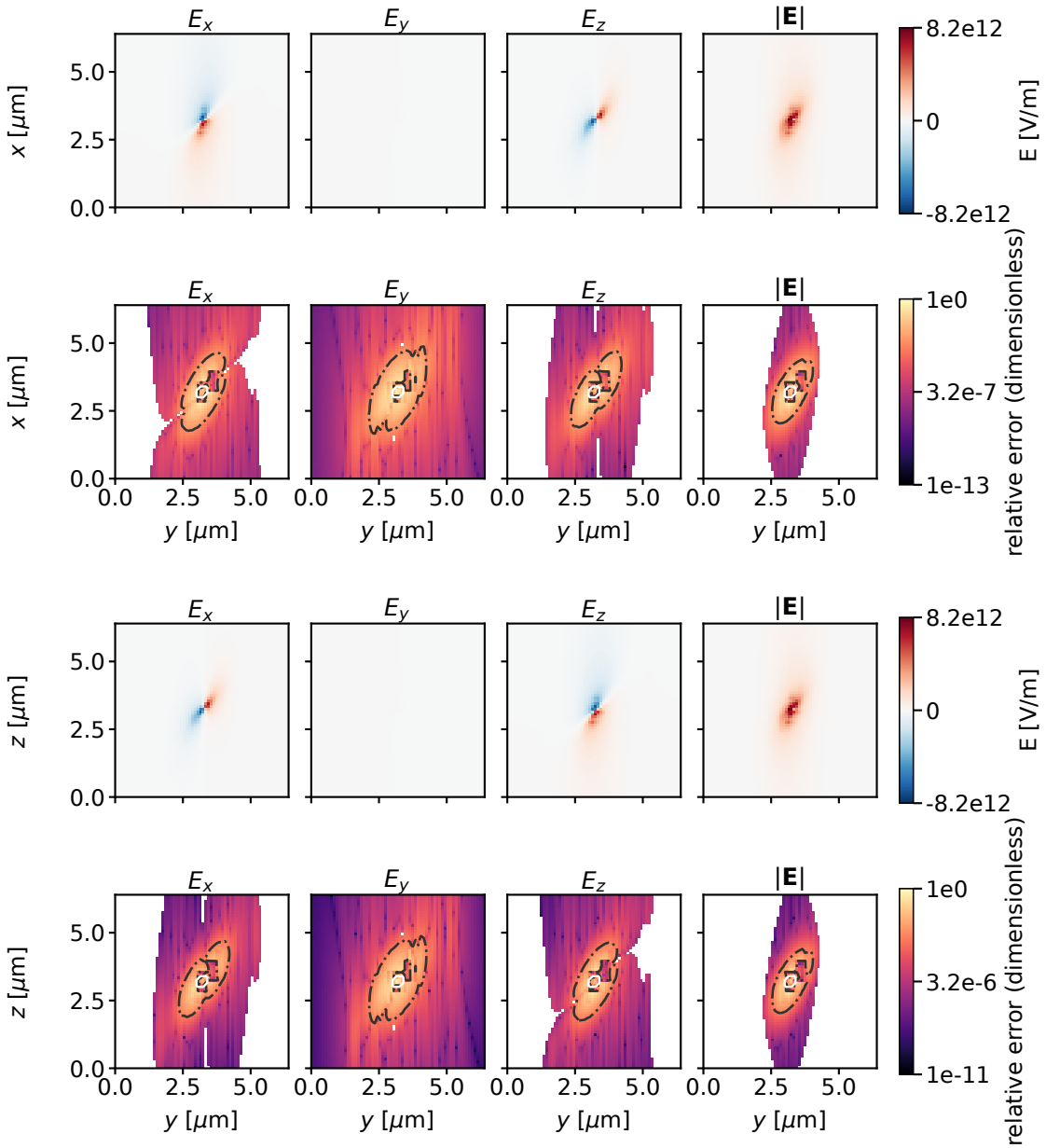


Figure 4: Forward-case electric validation ($N = 200$, $Q = -100\,\mathrm{pC}$, $\gamma = 10$ along $+y$, 64×64 planes, cell 10^{-7} m, `BDATA2`): x-y plane at $z = 3.2\,\mu\mathrm{m}$ (upper) and z-y plane at $x = 3.2\,\mu\mathrm{m}$ (lower), with y horizontal. Each image shows four columns $E_x, E_y, E_z, |E|$ with one shared linear scale and one colour bar per row; the lower row is logarithmic error, masked below 10^{-3} of the peak, and the white outline marks the TSC support. The in-support fast-versus-arbiter error against the order-12 `indfield` reference is 1.6% at the worst node.

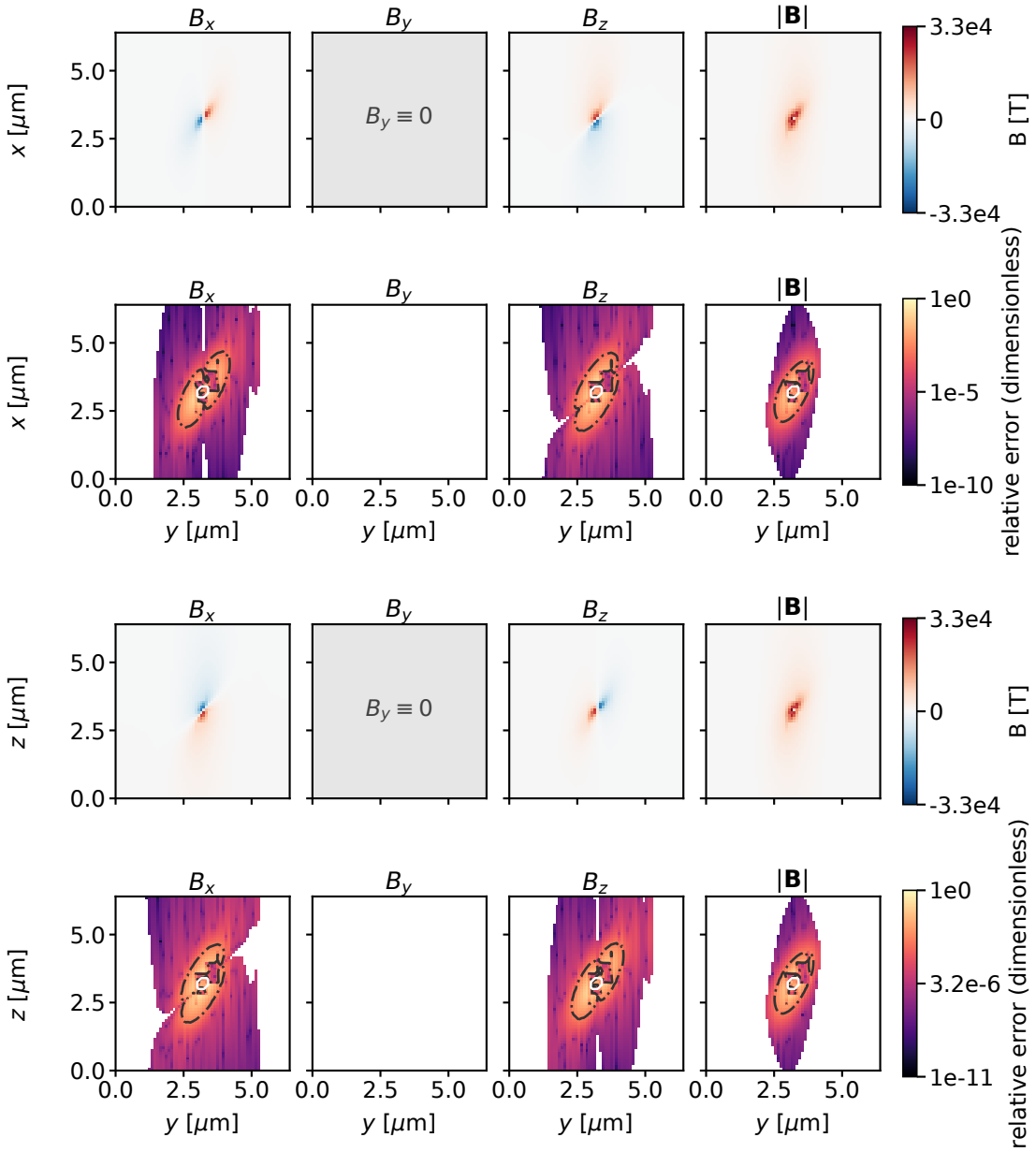


Figure 5: Magnetic companion to Fig. 4 with the same planes and layout, showing four columns $B_x, B_y, B_z, |B|$; $B_y \equiv 0$ for motion along $+y$ is marked in its panel. Each row has one shared linear field scale and one logarithmic error scale and colour bar, with the same TSC-support outline as the electric figure. The magnetic field is about 10^{-10} of the E-dominated local field at $\gamma = 10$.

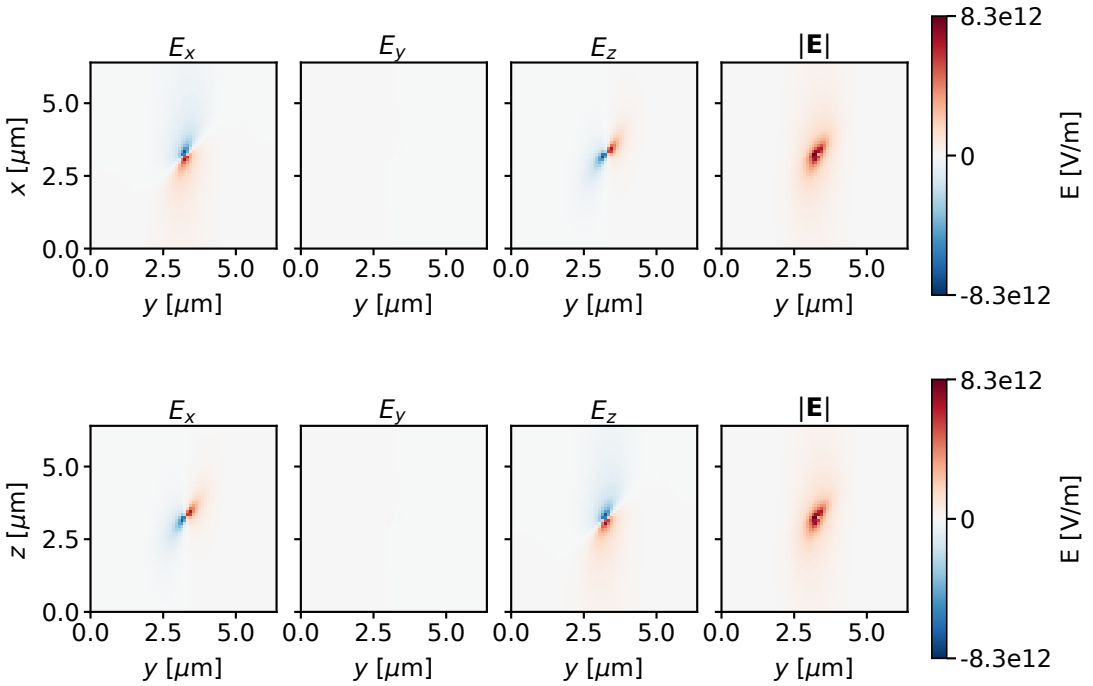


Figure 6: Fluctuation-case electric reference with the same geometry and bunch as Fig. 4 but a 1.045 mrad rms x-divergence (`BDATA2`), showing four columns $E_x, E_y, E_z, |E|$ from the order-12 `indfield` reference. Each row uses one shared linear scale and one colour bar. picBunch v0.9.0 rejects a momentum spread, so only the reference field is shown and the mesh panels are labelled unavailable.

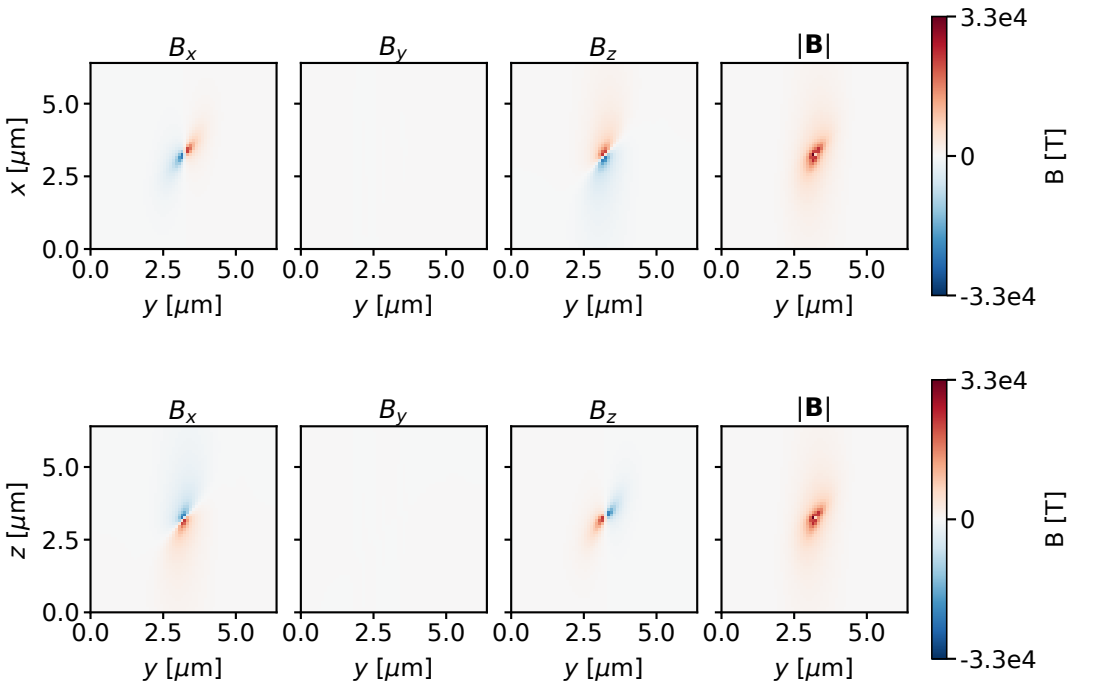


Figure 7: Magnetic reference for the fluctuation case, with the same planes, four columns $B_x, B_y, B_z, |B|$, and one shared scale and colour bar per row as Fig. 6. The x-divergence lifts B_y off zero, although it remains about 3×10^{-4} of the dominant component. Only reference fields from the order-12 `indfield` reference are shown; the picBunch panels are unavailable.

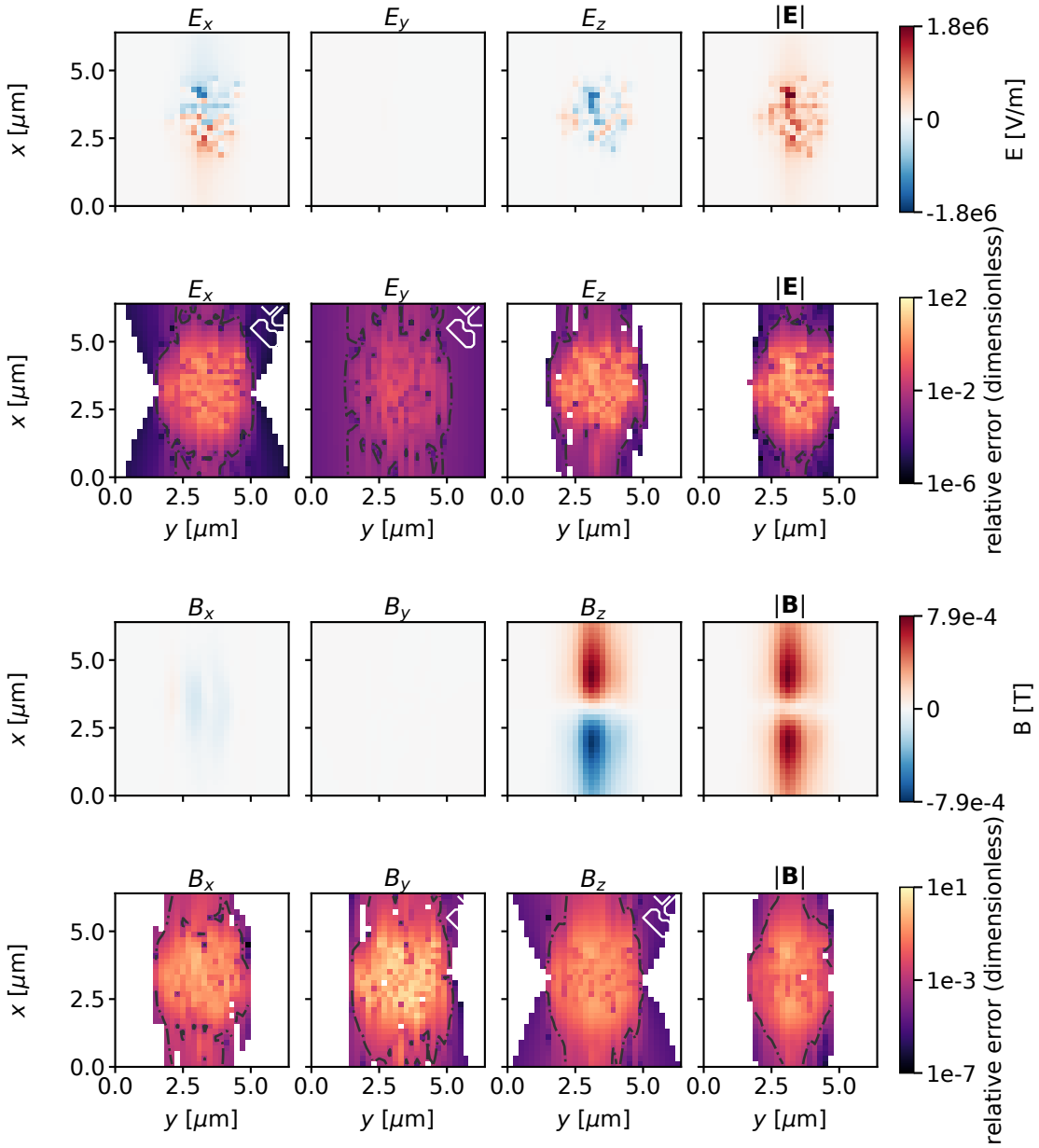


Figure 8: Binned 1 mrad case (`VAR1b`: $N = 200$, $Q = -32$ aC, $\gamma = 10$, 32×32 x-y plane at $z = 1.6$ µm, cell 2×10^{-7} m), with the picBunch velocity-binned field against the order-12 `indfield` reference. The upper image is electric and the lower magnetic, each with four columns, one shared scale and one colour bar per row; error rows are logarithmic and masked below 10^{-3} of the reference peak. The solid white outline marks the TSC support.

Table 10 lists the measured worst relative error per component and the fraction of unmasked plane pixels above 10^{-3} for the earlier $N = 1000$, $32 \times 32 \times 16$ set, with error-column pixels below 10^{-3} of the plane peak masked white. The new validation figures above supersede those slices for the forward and the 1 mrad cases.

The error is largest where these planes cut the bunch core: the spectral far-field convolution is not the dominant term there, and the maps show the measured error concentrated at the beamed core and falling away toward the plane edges. The far field is measured separately (Table 8, Fig. 3); the near-shell contribution is measured at the same geometry in Section 6.7.

6.6 The fast near evaluator, per component, by region

Table 11 gives the worst per-component relative error of the fast evaluator against the independent continuum, by region and by γ [fast-near worker, 2026]. The metric is the artifact formula $\max|\text{mesh} - \text{ref}|/(1 + \max|\text{ref}|)$ over 20 interior, 16 face and 16 outside points, cells 10^{-7} m. "Interior" is inside the TSC support, "face" is on its boundary, and "outside" is 1.6 to 3.6 cells away. The stationary ($\gamma = 1$) case is order 5; the moving axial case is order 8; the moving oblique interior is

Table 10: Mid-bunch slice statistics, mesh-only. Worst relative error per component and fraction of unmasked plane pixels above 10^{-3}, x-y plane at $z = 1.60\,\mu\text{m}$ and z-y plane at $x = 3.20\,\mu\text{m}$, $N = 1000$, $32 \times 32 \times 16$, cell $2 \times 10^{-7}\,\text{m}$, mesh against the independent exact TSC driver (FP64 quadrature, order 4, max level 5, tolerance 10^{-3}) [beam-direction worker, 2026, slice worker, 2026b]. The $\gamma = 10$ bunch moves along $+y$; y is the horizontal axis in every map. Node spacing $h = 1$ PIC cell; no near shell. The $\gamma = 1$ reference uses the archived tolerance 10^{-4} (driver floor 9.075×10^{-4}, 14 not converged).

plane	component	worst rel. err	frac $> 10^{-3}$
$\gamma = 10$			
x-y (z=1.60 µm)	E_x	1.81	54.1%
	E_y	1.16×10^{-1}	49.9%
	E_z	5.33	78.3%
	B_x	6.49×10^{-1}	72.7%
	B_z	3.77×10^{-1}	58.7%
z-y (x=3.20 µm)	E_x	5.06	90.4%
	E_y	1.17×10^{-1}	61.1%
	E_z	1.92	82.4%
	B_x	4.34×10^{-1}	80.9%
	B_z	7.16×10^{-1}	84.3%
$\gamma = 1$ *(stationary)*			
x-y (z=1.60 µm)	E_x	1.20×10^{-1}	22.8%
	E_y	1.34×10^{-1}	23.3%
	E_z	4.15×10^{-1}	74.7%
z-y (x=3.20 µm)	E_x	4.15×10^{-1}	49.6%
	E_y	1.37×10^{-1}	61.7%
	E_z	1.23×10^{-1}	53.9%

Table 11: Fast near evaluator, worst per-component relative error against the order-12 `indfield` reference, cells $10^{-7}\,\text{m}$, `jureca-h100` [fast-near worker, 2026, release worker, 2026]. Regions: interior (inside the TSC support), face (on its boundary), outside (1.6–3.6 cells). The "remainder order" column is the Gauss–Legendre order of the smooth remainder; the singular `asinh` box is O(1) in every row (moving axial uses the γ-stretched box, moving oblique the rest-frame octant box at order 8). Stationary and oblique remainder order 5, moving axial 8. The moving oblique interior is the measured limit of the fast backend.

γ, direction	remainder order	interior	face	outside
1, stationary	5	7.99×10^{-4}	1.96×10^{-5}	1.57×10^{-7}
10, axial	8	1.69×10^{-4}	1.11×10^{-4}	2.12×10^{-6}
10, oblique	5	8.15×10^{-2}	3.94×10^{-3}	1.01×10^{-4}

order 5. The oblique value is the measured limit of the fast backend: the rest-frame singular box of the oblique direction measures 8.15×10^{-2} at its default order [fast-near worker, 2026, release worker, 2026].

The per-component values behind the stationary interior row are 7.99×10^{-4}, 3.55×10^{-4}, 6.05×10^{-4}, 0, 0, 0 for $E_x, E_y, E_z, B_x, B_y, B_z$ (the magnetic field is identically zero for a stationary bunch); the moving axial interior row is 1.69×10^{-4}, 1.68×10^{-4}, 1.36×10^{-5}, 1.67×10^{-4}, 1.69×10^{-4}, 0; the moving oblique interior row is 2.43×10^{-2}, 7.42×10^{-2}, 6.32×10^{-2}, 8.15×10^{-2}, 2.96×10^{-2}, 3.89×10^{-2} [fast-near worker, 2026, release worker, 2026]. The stationary case meets 10^{-3} in every region; the moving axial case meets 10^{-3} in every region; the moving oblique interior is 8.15×10^{-2} [fast-near worker, 2026, release worker, 2026].

6.7 Near-shell accuracy and cost on the slice geometry

On the same $N = 1000$, $32 \times 32 \times 16$, cell $2 \times 10^{-7}\,\text{m}$ geometry, the exact near shell was run through the library path at the radius the envelope case assumed, $R = 3$ for $\gamma = 1$ and $R = 4$ for $\gamma = 10$, in mesh nodes of this $h = 1$ grid [slice worker, 2026a]. The mesh-only plane errors (Table 10) are 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$: these planes cut the sources, the region where the spectral convolution is not the dominant term. With the near shell the worst per-component plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$: at $R = 3/4$ on an $h = 1$ grid the shell radius covers 3/4 mesh nodes, which is a fraction of the TSC support extent (Section 6.5), and the measured residual is the mesh contribution outside the shell. The discrete Gauss-law indicator, the Yee backward-difference $\operatorname{div} E$ residual normalised to the peak deposited ρ/ε_0, moves from 0.3534 to 0.0895 for $\gamma = 1$ and from 5.026 to 0.399 for $\gamma = 10$, against reference values 0.0904 and 0.0814: $\gamma = 1$ reaches the reference value, $\gamma = 10$ drops 12.6× and stays 4.9× above it [slice worker, 2026a, worker, 2026f].

Table 12: Measured CLI wall at 2×10^6 particles, 20-cell bunch, $66 \times 68 \times 66$, cell 2.5×10^{-7} m, tile 4, one `H100` (`sm_90`) [release worker, 2026, fast-near worker, 2026]. "near-shell wall" excludes the mesh, "end-to-end wall" includes it. The mesh-only compute inside its 2.48 s wall is 0.116 s; the per-stage split (deposit, kernel, FFT, read-out, writer) is in the text.

configuration	near-shell wall (s)	end-to-end wall (s)	worst near error
mesh only	0	2.48	no near correction
support-only shell, $R = 1.5$, order 5	278.28	281.0	7.99×10^{-4}
three-cell shell, $R = 3$, order 5	908.14	955.4	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out
four-cell shell, $R = 4$, order 5	1508.12	1510.7	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out

The assembly itself is not the limiting term: the tests of Section 3.7 show the tiled subtraction exact to roundoff, so the residuals above are the mesh outside the shell [worker, 2026d].

Runtime at $N = 1000$ is 4.3 min for $\gamma = 1$ and 59 min for $\gamma = 10$ (one 16-thread socket) [slice worker, 2026a, near-shell worker, 2026]; the exact evaluator costs about 8× more per pair at $\gamma = 10$ than at $\gamma = 1$.

6.8 Cost at 2×10^6 particles, one H100

Table 12 gives the measured CLI wall times at 2×10^6 particles in a 20-cell bunch, grid $66 \times 68 \times 66$, cell 2.5×10^{-7} m, near-shell tile 4, on one `H100` (`jrc0880`, `sm_90`), with the achieved near error for each configuration [release worker, 2026, fast-near worker, 2026]. The near shell is the dominant term in every shell row; the mesh-only wall is 2.48 s, of which the compute is 0.116 s (deposit 34.8–40.5 ms, kernel 1.1 ms, FFT 7.3–7.6 ms, read-out 8.0–8.5 ms, writer 48.8–254.5 ms). The near evaluator is the shipped fast backend of Section 3.7, measured through the CLI, not as a standalone kernel. Pair counts are quoted in source-node pairs and in components: support 4.319×10^8 / 2.59×10^9, three-cell 1.458×10^9 / 8.75×10^9, four-cell 2.662×10^9 / 1.60×10^{10}, the neighbour counts of the 20-cell geometry [closed-form worker, 2026].

The support-only shell costs 281.0 s end-to-end at 7.99×10^{-4} interior error, the three-cell shell 955.4 s, and the four-cell shell 1510.7 s, at the same interior error [release worker, 2026]. The shell radius is the dominant cost lever: going from $R = 1.5$ to $R = 4$ multiplies the end-to-end wall by 5.4× at fixed particle count and grid, while the accuracy of the in-shell region is already at 7.99×10^{-4} at $R = 1.5$. The shell rate on the integrated path is 1.55×10^6, 1.61×10^6 and 1.76×10^6 pairs/s (source-node pairs) for the three radii, about 2× slower per component than the standalone kernel, the overhead of the per-tile local FFT and launch in the integrated path [release worker, 2026]. Fig. 9 places the three evaluated regions (TSC support, near shell, mesh) around one source and shows the measured error against distance for the $\gamma = 1$ and $\gamma = 10$ axial cases.

6.9 Device backend for the fast near evaluator

`v0.9.1` carries a device backend, `-near-shell fast-gpu`: the same algebra as the host fast path, one thread per (pair, component), on the CUDA device [fast-near worker, 2026, release worker, 2026]. Against the host fast path it agrees to 5.4×10^{-16} to 8.0×10^{-16}; when no CUDA device is present the run falls back to the host path with a warning, so the flag is safe on host-only builds [fast-near worker, 2026, release worker, 2026]. The device backend is the fast path of Section 3.7 moved to the device; the lid slice run, one near-shell tile of the shipped problem, is 14.13 s on the device versus 25.68 s on the host, a 1.82× speed-up for that stage [fast-near worker, 2026, release worker, 2026].

6.10 Distributed FFT

A distributed FFT (`experimental/distributed-fft`, `d602484d`) decomposes the output field in z slabs, one slab per rank, and moves the z rows of each 2-D slice between ranks with `MPI_Alltoallv` transposes around each 2-D transform [distributed FFT worker, 2026, mpi worker, 2026a]. Against the replicated path it agrees to 2.2×10^{-16}, and the written outputs are bitwise equal at 1, 2 and 4 ranks, so the decomposition changes no value [distributed FFT worker, 2026, mpi worker, 2026a]. At a 640^3 output the per-rank host memory is 41.1 GB at 2 ranks and 20.6 GB at 4, the quantity the decomposition buys: the output field is slab-local, while the bunch remains replicated on every rank [distributed FFT worker, 2026, mpi worker, 2026a]. The transposes are 74–85% of the FFT-plus- transpose time, so the distributed path is a memory feature rather than a speed feature [distributed FFT worker, 2026, mpi worker, 2026a].

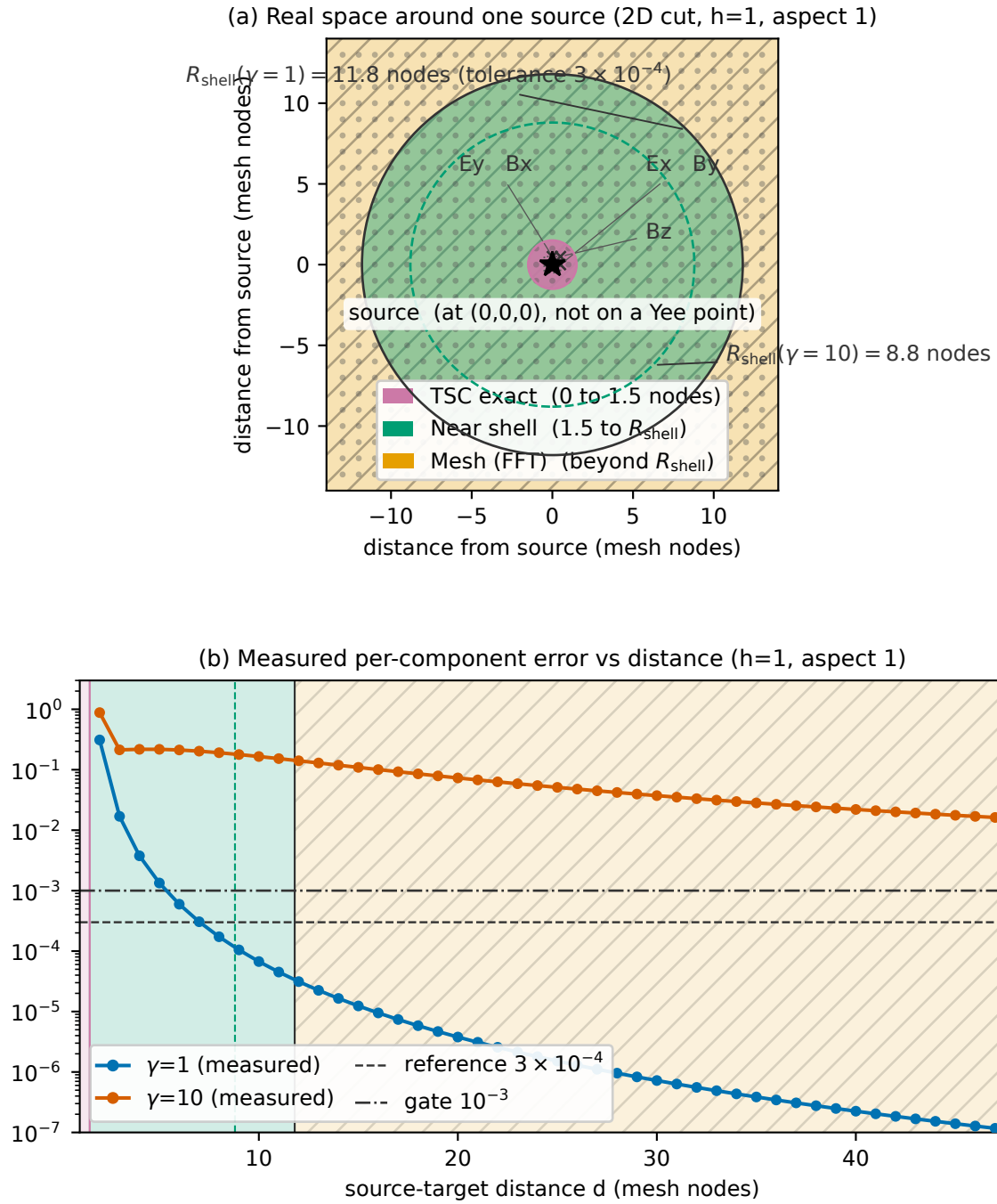


Figure 9: Where the algorithm evaluates the field: real space around one source (top) and the measured error against distance (bottom) at $h = 1$, aspect 1, `rosi_v100` [scaling worker, 2026a, schematic worker, 2026]. Shaded regions are the TSC support (to 1.5 nodes), the near shell (to R_{shell}), and the mesh beyond, with $R_{\text{shell}} = 11.8$ nodes for $\gamma = 1$ and 8.8 nodes for $\gamma = 10$ at the 3×10^{-4} tolerance. The $\gamma = 1$ error crosses 3×10^{-4} at $d = 8$ nodes, while the $\gamma = 10$ curve stays above it to $d = 48$.

6.11 The disagreement is per-source

On a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$), the disagreement with the exact-quadrature reference does not grow with particle count: the band is flat in N from 1 to 1000 (10^{-3}–3×10^{-2}) [worker, 2026g]. The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.415–0.65 cells from a source, inside that source's TSC support; a charge-weighting defect would grow with N, and it does not [worker, 2026g].

Table 13: Single-particle mesh error at $d = 8$ cells, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca-h100` [worker, 2026c]. Peak-normalised absolute error, worst per component, at $h = 1$ and $h = 1/4$. The error grows with γ; at $h = 1$ the $\gamma = 50$ and $\gamma = 100$ values exceed the peak field, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises.

γ	$h = 1$	$h = 1/4$	slope p ($h = 1 \to h = 1/4$)	note
10	4.4×10^{-2}	6.9×10^{-5}	1.85	
50	1.05	5.6×10^{-2}	1.06	$h = 1$ off by more than the peak
100	5.47	0.82	1.01	$h = 1$ off by more than the peak

6.12 Anisotropic cells: the gap is the TSC convolution

At anisotropic spacing the accuracy is quantified per geometry, not inherited from the cubic case [worker, 2026h,i]. On a grid with per-axis cell spacing $(1, 1, 4)$ the stretched z axis matches the continuum to $\approx 10^{-3}$–10^{-7} across the in-grid distances, the same scale as the cubic control; the short x/y axes sit in the near-field TSC regime at a short physical distance ($d = 4$ cells is only 4 m along x/y, so $E_z \approx 0.19$ stationary and E_x/E_y up to ≈ 0.9 at $d = 4$ for the $\gamma = 10$ case) [worker, 2026i]. The mesh is not wrong because the spacings differ; the near points on the short axes are physically closer to the source. A reported triplet for this geometry, "the exact TSC convolution is 0.9638× the point Coulomb field (3.2% offset) and the mesh matches the continuum to 0.44% at 12 cells," does not appear in `A1b/RESULTS.md` or `A1c/RESULTS.md` and is not repeated here as a measured value [worker, 2026h,i].

The h^3 factor. The sampled convolution carries the Riemann volume of the continuum integral of Eq. (1): each sample covers h^3 cells, so the read-out applies $h_x h_y h_z/(M_x M_y M_z)$ (Section 3) [extraction worker, 2026].

6.13 The γ landscape: 10 to 100

The error landscape for a single particle at $\gamma = 10, 50, 100$ is measured on a 96-cell box, cell 10^{-7} m, aspect 1, at $h = 1$ and $h = 1/4$, axial and oblique (45°), against the independent continuum [worker, 2026c]. The metric is the peak-normalised absolute error $\max|\text{mesh} - \text{ref}|/\max|\text{ref}|$ per component, at $d = 8$ cells for the h-comparison. These runs used the $v0.5.0$ tree at padding 1 (padding 2 was infeasible for the 384^3 $h = 1/4$ grid); the $d = 48$ rows carry a small padding-1 periodic-wraparound floor of order 10^{-4} to 10^{-3}, so the $d = 48$ values are not padding-clean [worker, 2026c].

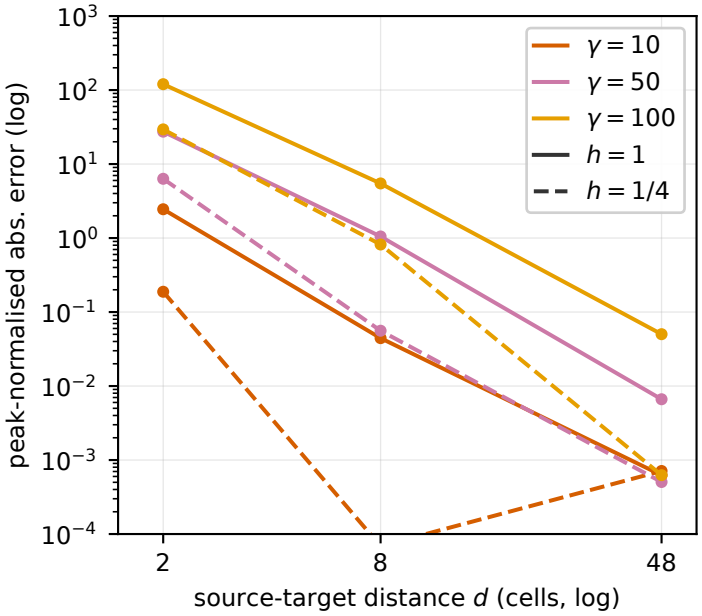


Figure 10: Peak-normalised absolute error against distance on log-log axes, single particle, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca-h100` [worker, 2026c]. Solid/dashed is $h = 1/1/4$ and hue is $\gamma = 10$ (vermillion), 50 (purple), 100 (orange). At $h = 1$ the error exceeds the peak field by $d = 8$ cells for $\gamma = 50$ and $\gamma = 100$.

Table 13 is the measured behaviour, and Fig. 10 shows the same quantities against distance. The error grows steeply with γ: 4.4×10^{-2} at $\gamma = 10$, 1.05 at $\gamma = 50$, 5.47 at $\gamma = 100$, at $h = 1$ and $d = 8$ cells [worker, 2026c]. Refining to $h = 1/4$ brings these to 6.9×10^{-5}, 5.6×10^{-2}, and 0.82. The controlling length is the beaming cone width d/γ: for $\gamma \geq 50$ the

cone is sub-cell, the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) collapses to about 1, and refinement stops paying off [worker, 2026c]. The h targets for 10^{-2} derive from the dominant $d = 2$ near-charge band (the error $\sim h^p$ fit uses the $d = 2$ band, not the $d = 8$ rows in Table 13). The oblique (45°) target is $h \leq 0.002$ at $\gamma = 50$; the axial (0°) target is $h \leq 0.001$ at $\gamma = 50$ (5.6×10^8 times the nodes). For $\gamma \geq 50$ the cone is sub-cell and 10^{-2} needs $h \leq 0.002$ ($\geq 10^8$ times the nodes); 10^{-3} is out of reach ($> 10^9$ to 10^{12} times the nodes) [worker, 2026c]. A shell of radius of order γ cells is what the near-charge error requires at $\gamma = 50$ to 100 [worker, 2026c]. These are measured limits of the mesh at a fixed distance in cells, stated as the behaviour the data show.

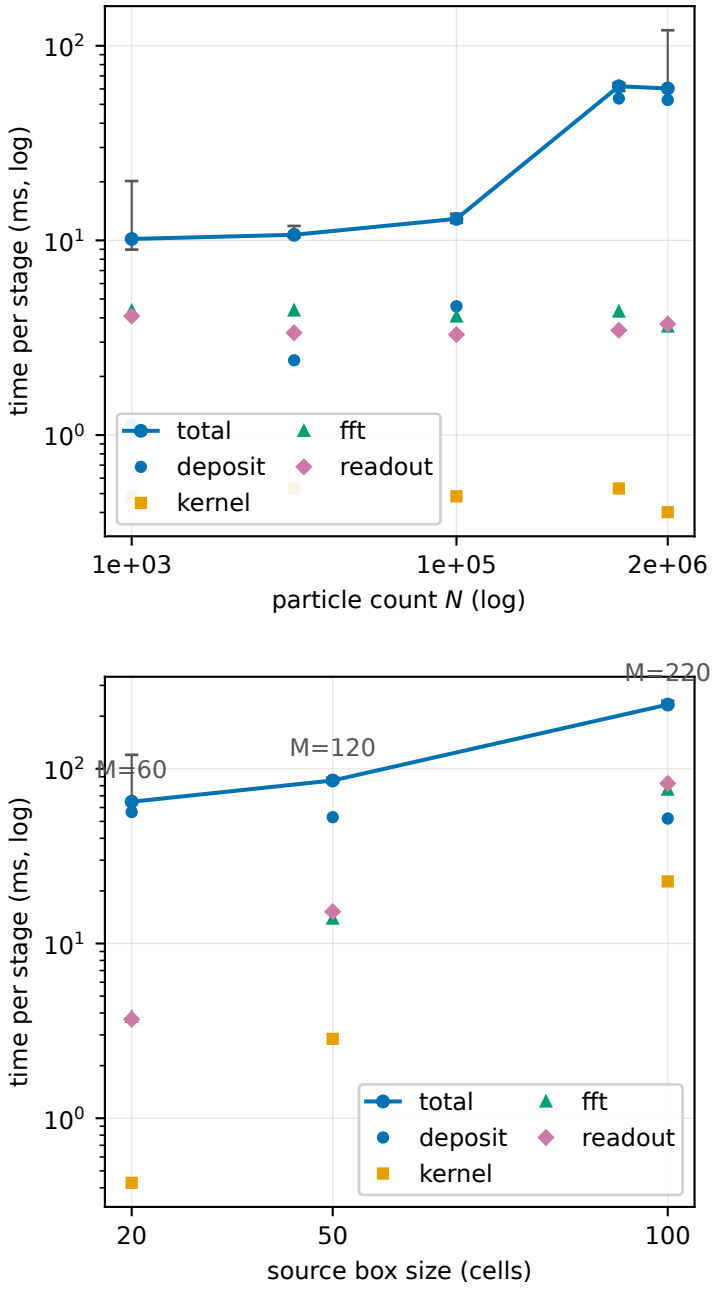


Figure 1: Pipeline time versus particle count (top, log-log) and source-box size (bottom, semi-log), padded array $M = 60/120/220$ cells, $\gamma = 10$, $h = 1$, double precision, `rosi_v100`/CUDA 12.6 from `SB1`. Blue is the total and markers the per-stage medians; the deposit dominates at $N = 2 \times 10^6$ (57 of 64.8 ms), and the total rises from 64.8 to 233 ms as M grows from 60 to 220.

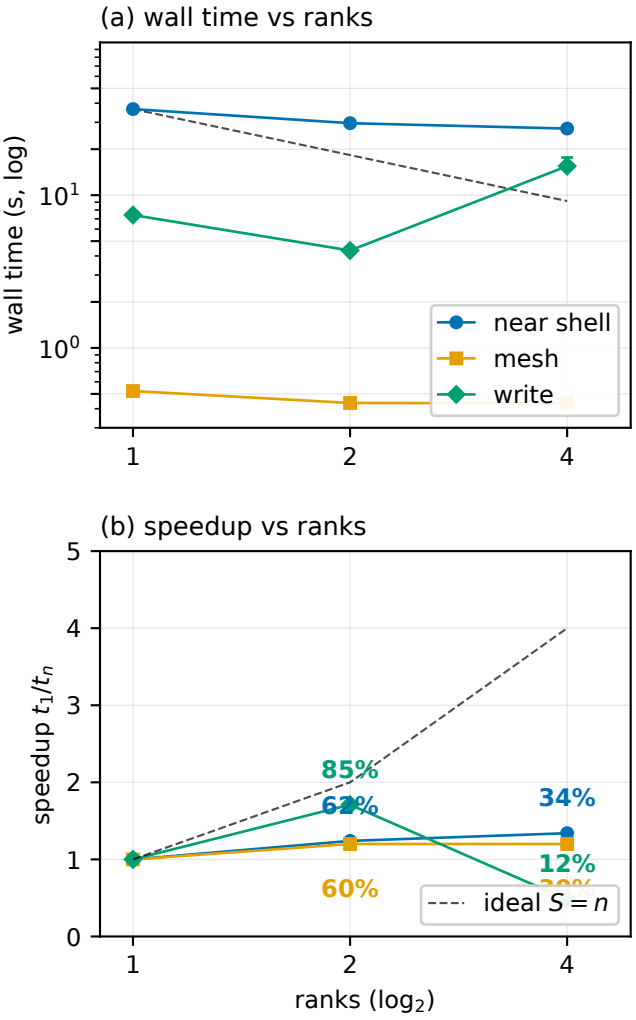


Figure 2: Strong scaling of the three rank-parallel pieces on `rosi_v100` (idle window, medians of three runs) for 1–4 ranks, with dashed ideal $1/n$ and $S = n$ references. The near shell reaches 1.34×, the component-parallel mesh 1.20×, and the collective writer 1.71× at two ranks then 0.48× at four.

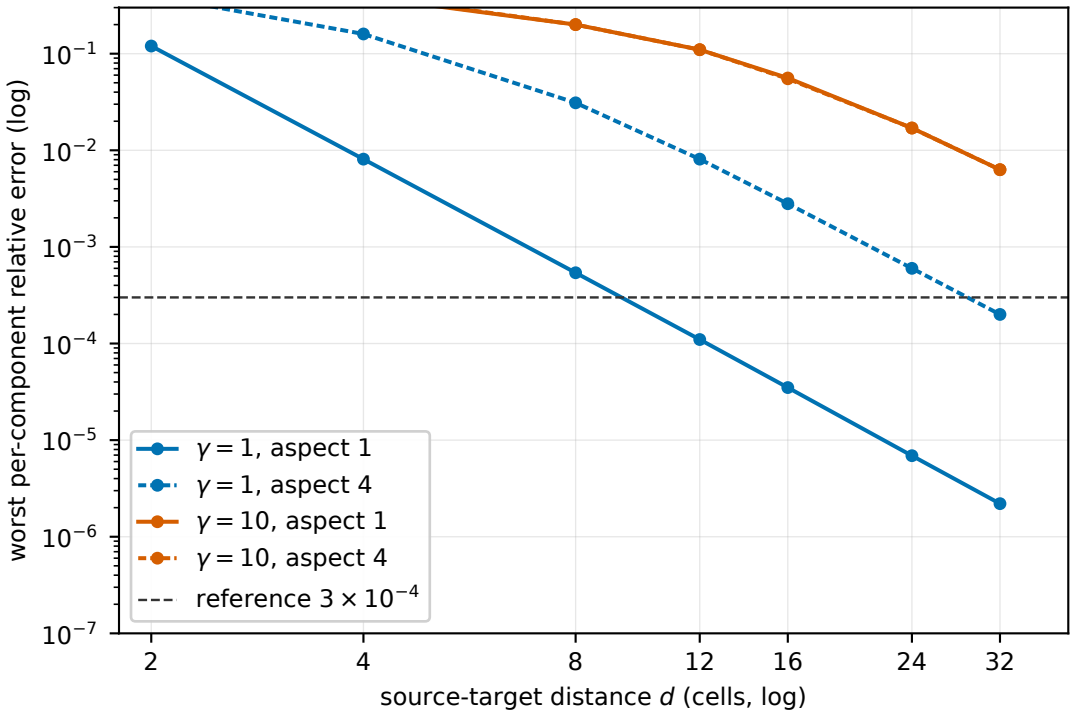


Figure 3: Worst per-component relative error against source-target distance d for the single-particle floored electric series on `rosi_v100`, both axes logarithmic; hue is γ (blue $\gamma = 1$, vermillion $\gamma = 10$) and line style is aspect (solid 1, dashed 4), against the order-12 `indfield` reference (refine 3) [worker, 2026e]. The error falls from 1.2×10^{-1} at $d = 2$ to 2.2×10^{-6} at $d = 32$ for $\gamma = 1$, aspect 1.

Fig. 11 shows the $\gamma = 10$ field and error on a $128 \times 128 \times 64$ domain, cell 2×10^{-7} m, $N = 100$, against the independent continuum at tolerance 10^{-3} [large-domain worker, 2026]; Fig. 12 is its magnetic companion, and Fig. 13 is the stationary-bunch electric case. The boundary field is about 10.3% of the peak (worst per-component, $\gamma = 10$ z-y E_z), so the colour scale is set by the bunch and the far decay is visible inside the frame. The core error is the method error of the mesh where the planes cut the bunch, far above the 10^{-3} reference floor [large-domain worker, 2026].

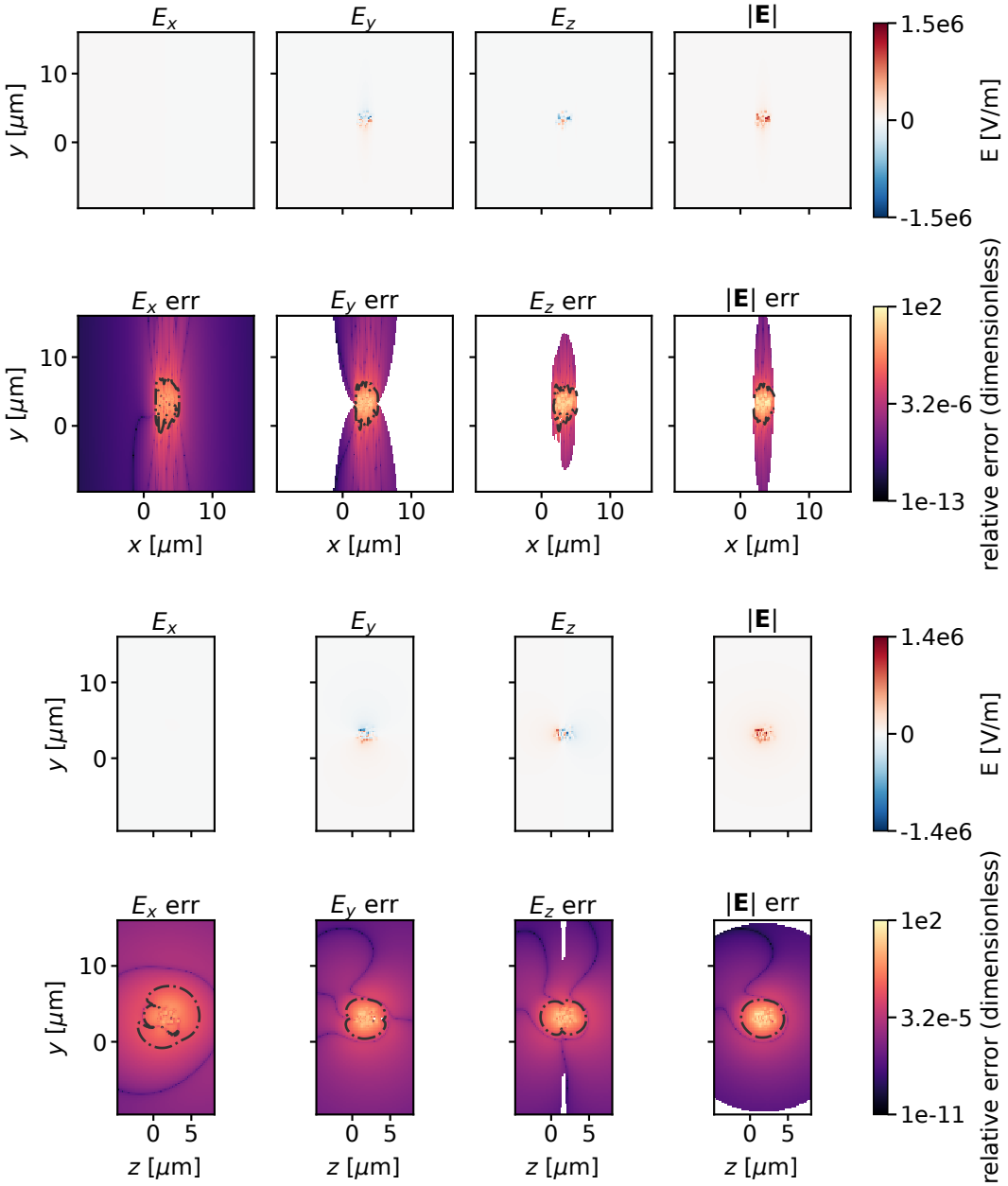


Figure 11: Large-domain electric slices ($\gamma = 10$, $128 \times 128 \times 64$, cell 2×10^{-7} m, $N = 100$, `jureca-h100`) from the earlier axial-x data set (not the $+y$ beam-axis set); x is horizontal in the x-y cut and z is horizontal in the z-y cut. Upper and lower images are x-y and z-y planes, each showing four columns $E_x, E_y, E_z, |E|$ with one shared linear scale and one colour bar per row above a logarithmic error row against the order-4 driver reference. The boundary field is 10.3% of the peak (worst component, z-y E_z), so the colour scale is set by the bunch.

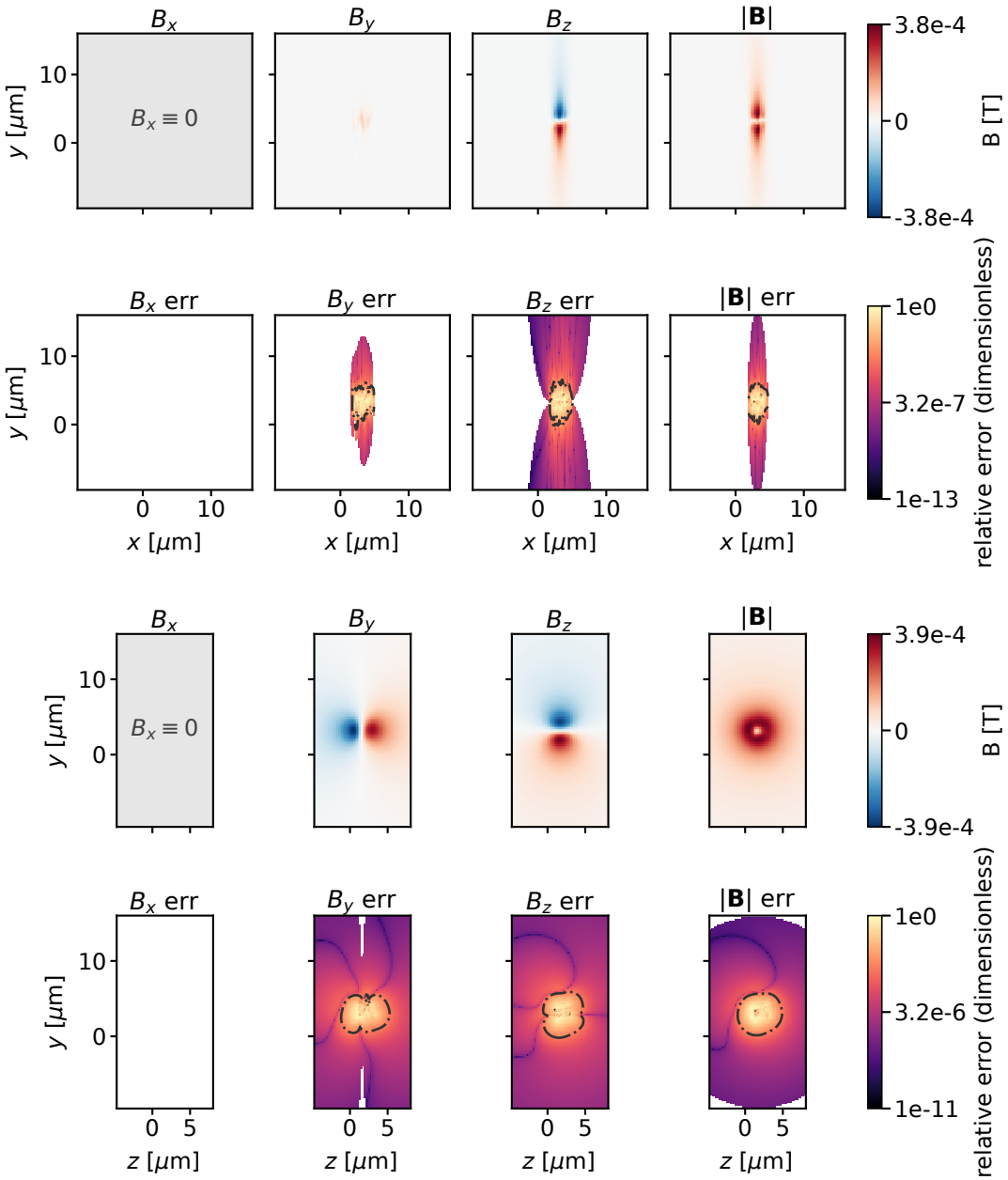


Figure 12: Magnetic companion to Fig. 11 at $\gamma = 10$, same planes and earlier axial-x geometry; x is horizontal in the x-y cut and z is horizontal in the z-y cut, and $B_x \equiv 0$ is marked in its panel. Each image shows four columns $B_x, B_y, B_z, |B|$ with one shared linear scale and one colour bar per row, and the same logarithmic error convention as the electric figure. The plane peak is 6.2×10^{-4} T.

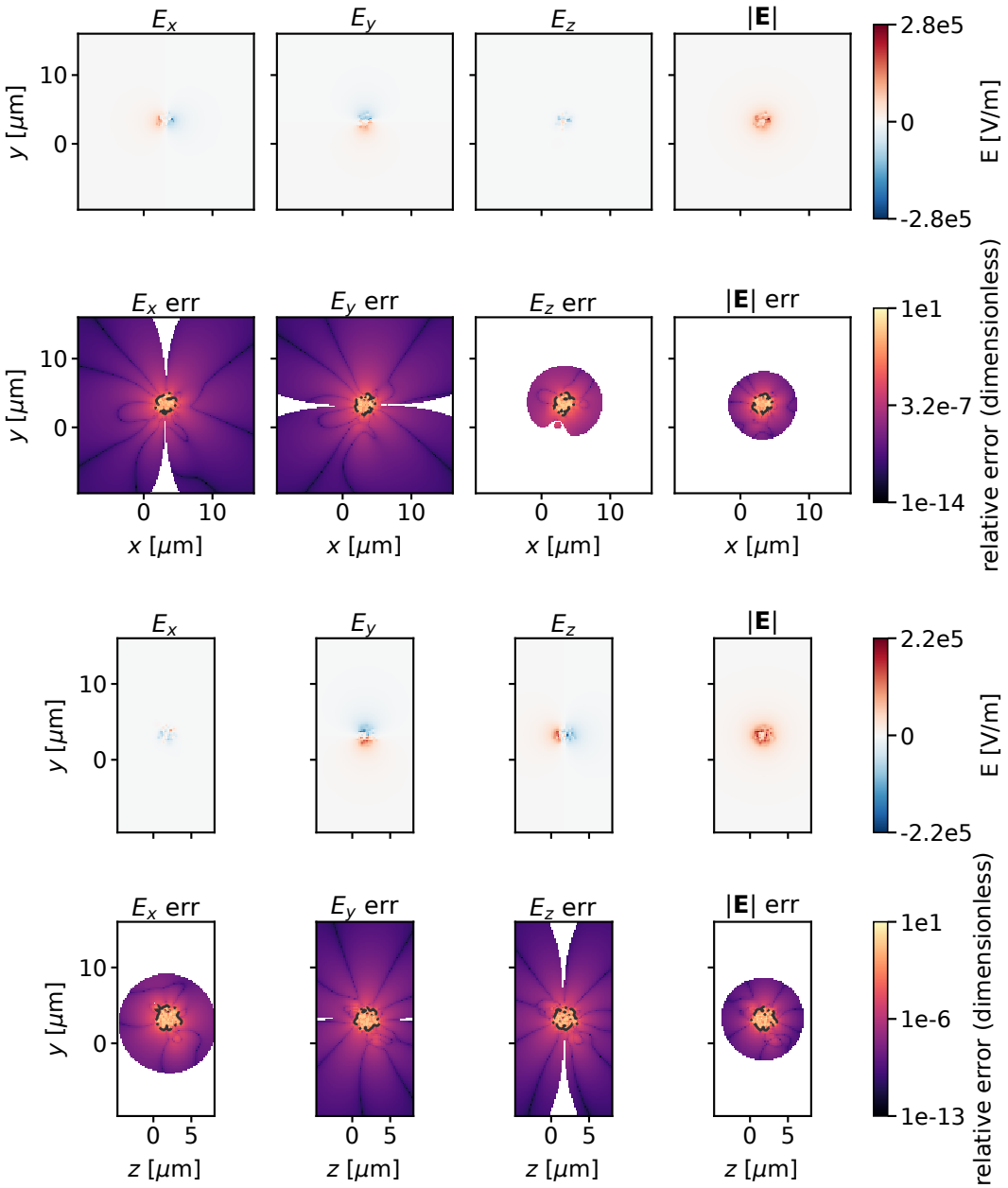


Figure 13: Large-domain electric slices for the stationary bunch ($\gamma = 1$), same domain and layout as Fig. 11 [large-domain worker, 2026]; x is horizontal in the x-y cut and z is horizontal in the z-y cut. Each image shows four columns $E_x, E_y, E_z, |E|$ with one shared linear scale and one colour bar per row, above a logarithmic error row. The core error reaches 7.6×10^{-1} (z-y E_x), far above the 10^{-3} reference floor.

6.14 Six-velocity single-particle validation

Figures 14, 16, 18, 15, 17 and 19 show the shipped mesh field for one electron on a 64^3 grid, cell 10^{-7} m, against the independent order-12 continuum reference on the two mid planes. The six rows are the velocity cases $\beta = 0.1$, $\beta = 0.5$, $\gamma = 1$, $\gamma = 10$, $\gamma = 30$ and $\gamma = 100$, with y horizontal and increasing left to right. The measured worst relative errors are 3.49×10^{-3}, 3.73×10^{-3}, 3.47×10^{-3}, 1.20×10^{-1}, 7.39×10^{-1} and 3.21 in that order. The error grows with γ because the field is compressed into a fraction of a cell: at $\gamma = 30$ and $\gamma = 100$ the Lorentz pancake is thinner than the mesh spacing, so those rows show the geometric sub-cell limit of the mesh, not a solver defect.

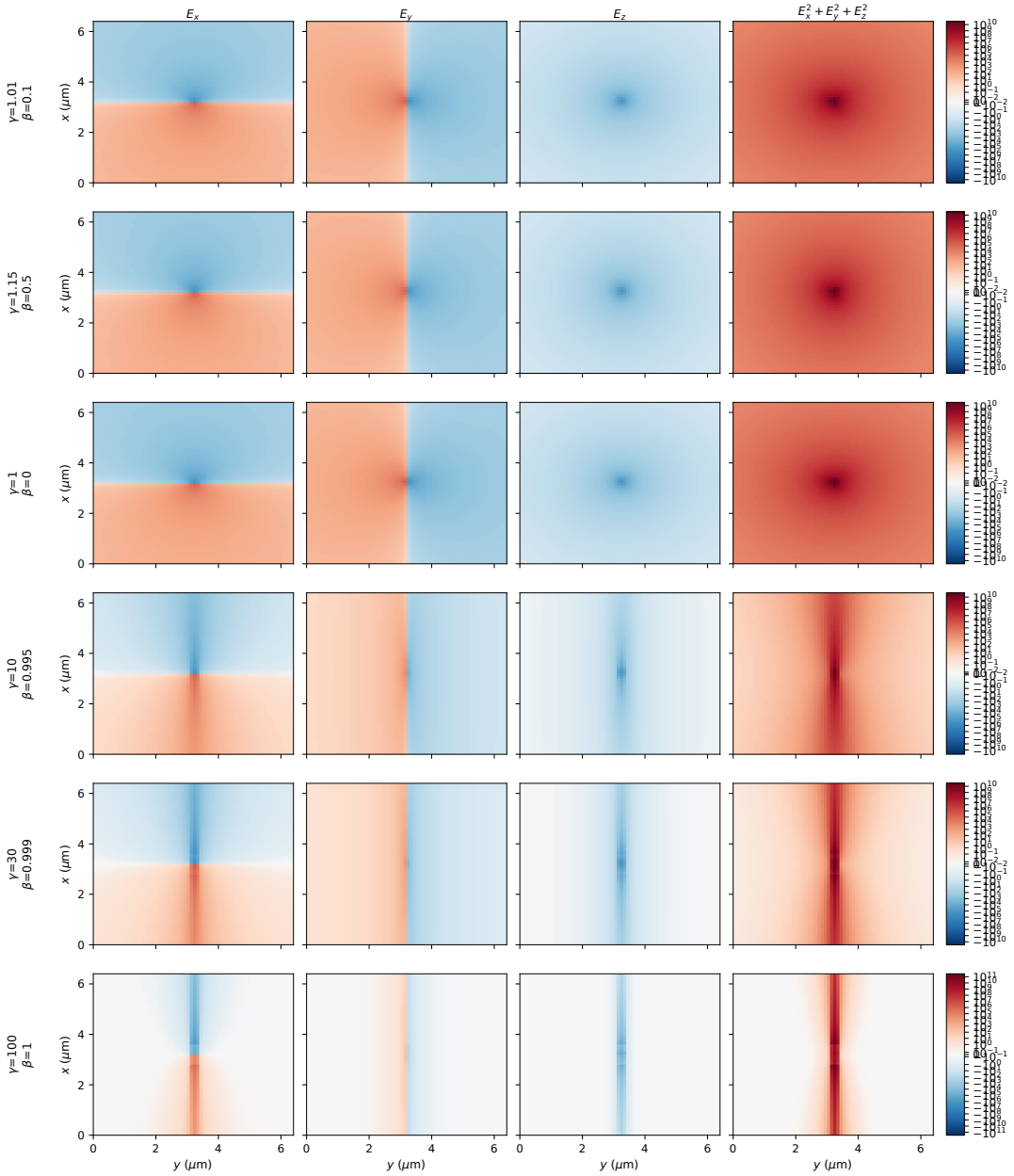


Figure 14: Single-electron electric field on the x-y plane at mid-z for the six velocity cases $\beta = 0.1$, $\beta = 0.5$, $\gamma = 1$, $\gamma = 10$, $\gamma = 30$ and $\gamma = 100$ (rows). The columns are E_x, E_y, E_z and $E_x^2 + E_y^2 + E_z^2$; each row has one shared symmetric-log colour scale and one colour bar, and y increases left to right in μm. The measured worst relative error over the six rows is 3.21 at $\gamma = 100$.

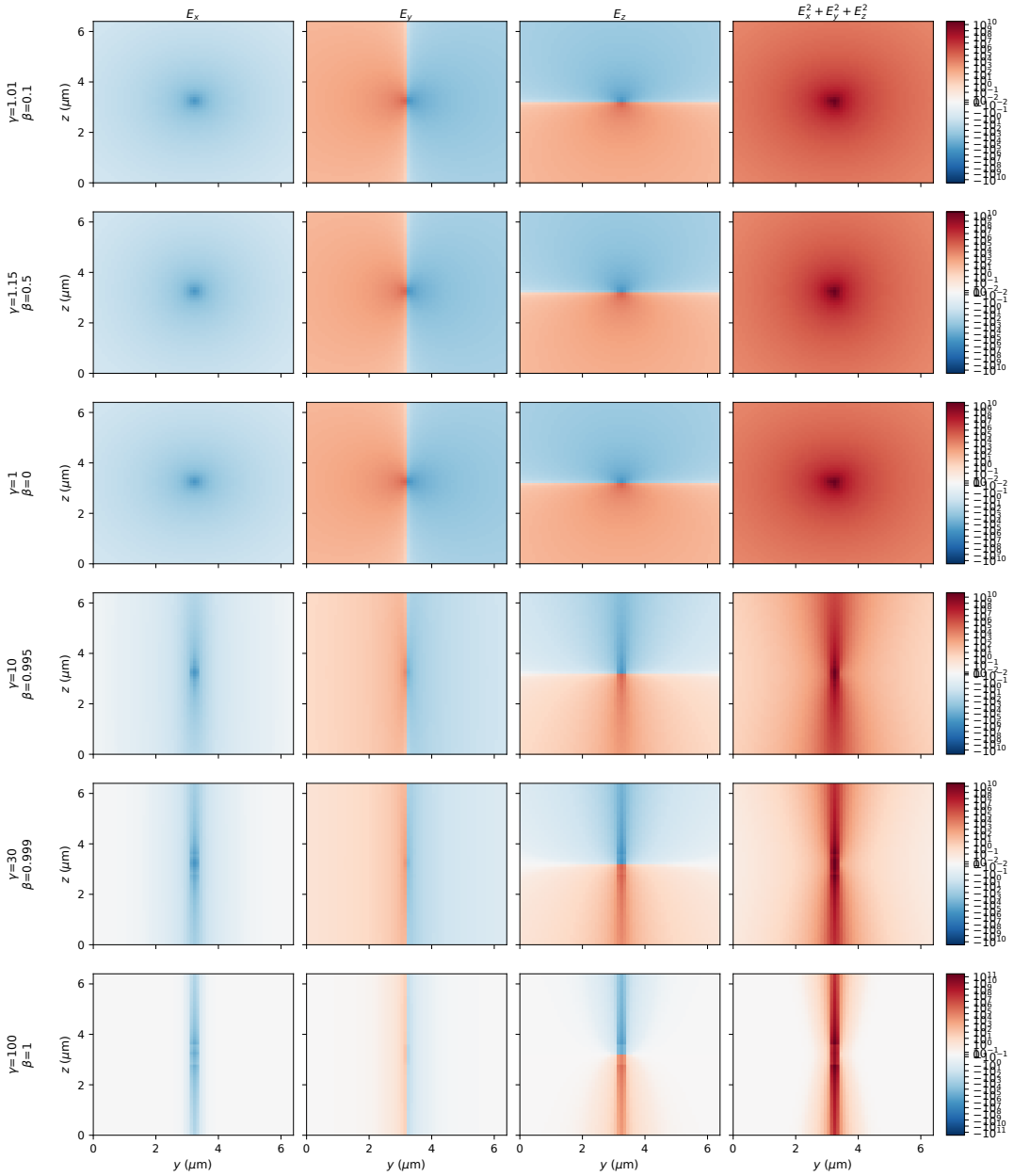


Figure 15: Same six-velocity electric data as Fig. 14 on the z-y plane at mid-x, with y horizontal and z vertical, four columns E_x, E_y, E_z and $E_x^2+E_y^2+E_z^2$, and one shared colour scale and one colour bar per row. The measured worst relative error is again 3.21 at $\gamma = 100$.

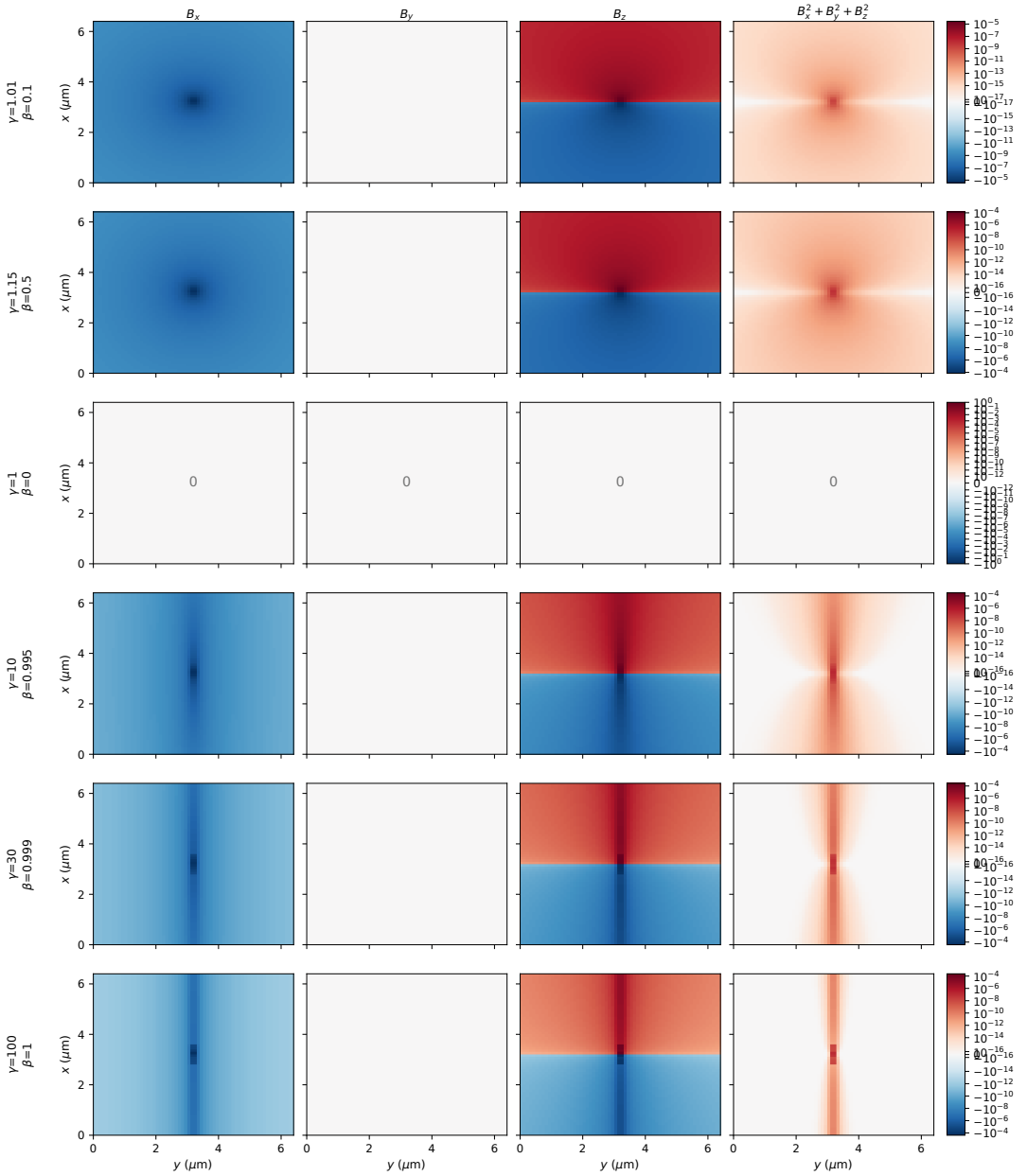


Figure 16: Magnetic companion to Fig. 14: x-y plane, six velocity rows, columns B_x, B_y, B_z and $B_x^2 + B_y^2 + B_z^2$, one shared symmetric-log colour scale and one colour bar per row. The field is identically zero at $\gamma = 1$, and the measured worst magnetic relative error over the six rows is 0.694 at $\gamma = 100$.

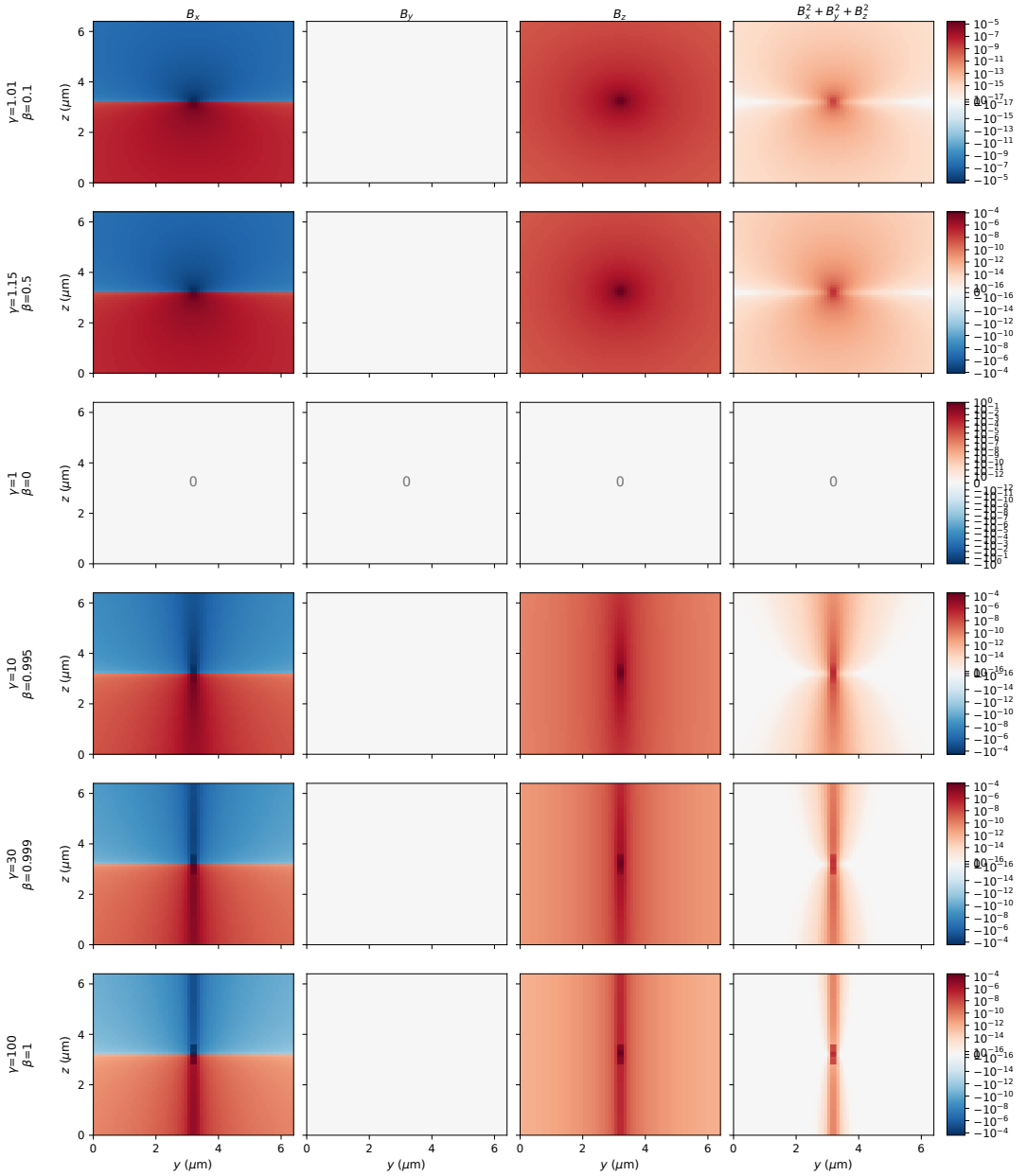


Figure 17: Magnetic companion on the z-y plane at mid-x, same six velocity rows and columns B_x, B_y, B_z and $B_x^2 + B_y^2 + B_z^2$, each row on one shared colour scale with one colour bar. The field is identically zero at $\gamma = 1$, and the measured worst magnetic relative error is 0.694 at $\gamma = 100$.

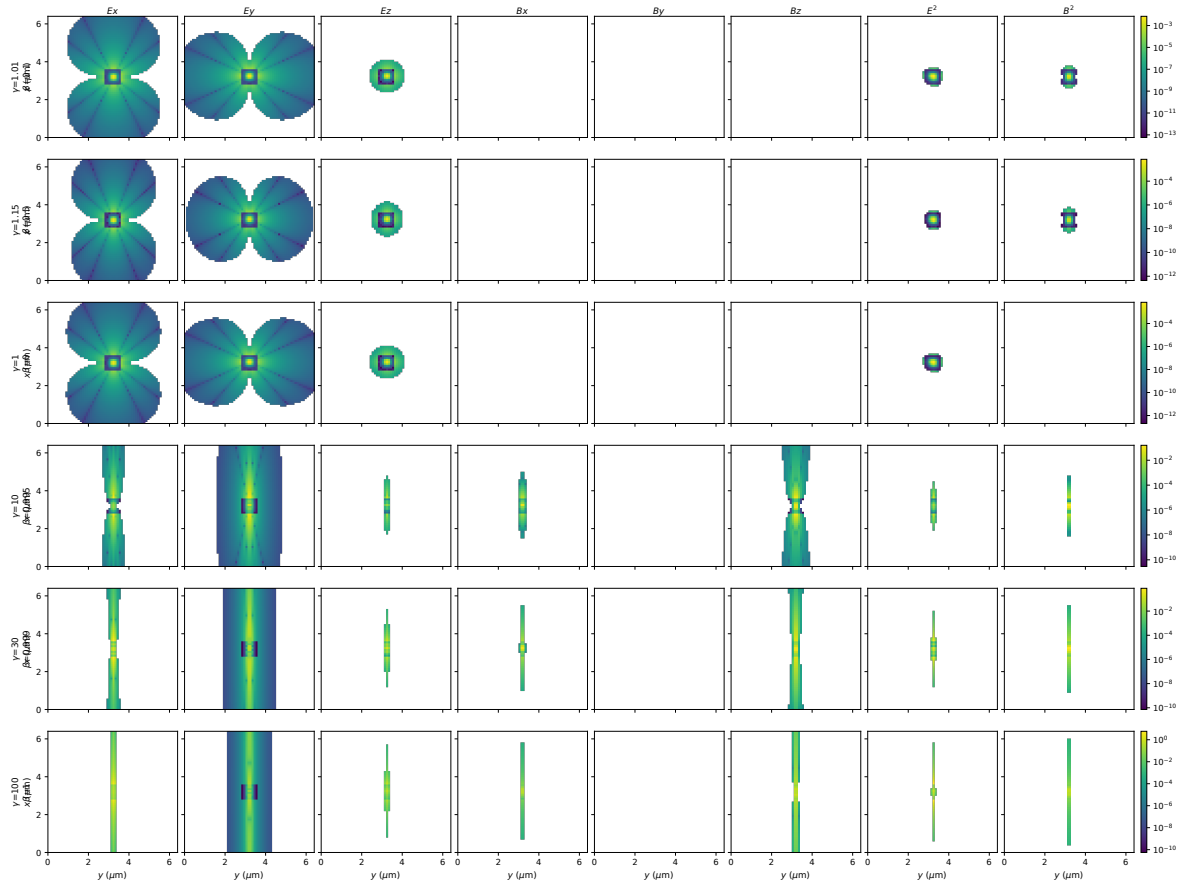


Figure 18: Relative error $|\text{mesh} - \text{ref}|/\text{peak(ref)}$ on the x-y plane for the six velocity rows, with columns E_x, E_y, E_z, E^2, B_x, B_y, B_z and B^2. Each row is logarithmic, has one colour bar, and is masked where the reference is below 10^{-3} of its own peak. The worst component error is 3.21 at $\gamma = 100$.

Table 14: Binned multi-velocity peaks, 64^3 grid, 200 sources, $\gamma = 10$, picBunch `v0.9.1` [multi-velocity worker, 2026b].

spread	$B{=}16$ $\lvert B_z\rvert$ (T)	$B{=}64$ $\lvert B_z\rvert$ (T)	$B{=}16$ $\lvert B_y\rvert$ (T)	$B{=}64$ $\lvert B_y\rvert$ (T)
energy 5%	2.32050×10^4	2.32049×10^4	0	0
angular 5°	2.48202×10^4	2.48213×10^4	1.2948×10^3	1.2955×10^3

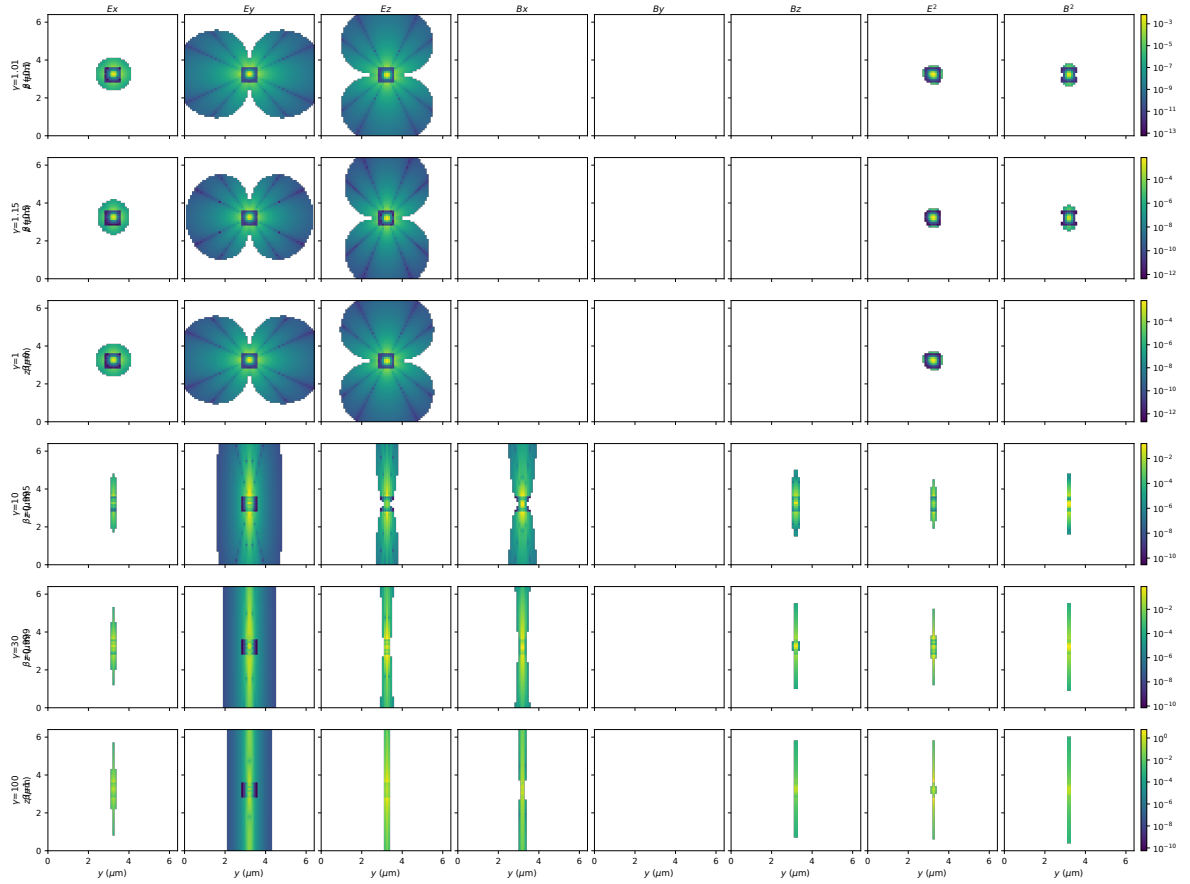


Figure 19: Same error convention on the z-y plane at mid-x, with the six velocity rows and the same eight columns. Each row is logarithmic, has one colour bar, and is masked where the reference is below 10^{-3} of its own peak. The worst component error is 3.21 at $\gamma = 100$.

6.15 Determinism

Repeated runs of the shipped CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes) [worker, 2026e]. The CLI also rejects a below-floor `-tolerance` (10^{-15}) with a nonzero exit [worker, 2026e]. The `gridUnitSI` is applied once, in the reader (`OpenPMD.cpp`), and the mesh geometry follows `gridSpacing` · `gridUnitSI`; a mesh without a positive `gridUnitSI` is rejected [extraction worker, 2026].

6.16 Binned multi-velocity binning, measured

The single-velocity scope of Section 2 is extended by splitting the velocity distribution into B classes and running one convolution per class; the near shell needs no binning, because each pair carries its own velocity [multi-velocity worker, 2026a,b]. The measurements below are on a 64^3 grid with 200 sources on `hal` (NVIDIA A30, `sm_80`, CUDA 13.4.92), picBunch `v0.9.1` (commit `c0bf82c`, tag `262927ba`) [multi-velocity worker, 2026b, integration worker, 2026a].

Single-velocity sanity. With zero spread the binned path is bitwise equal to the unbinned path: $B = 1$ and $B = 16$ give bit-identical fields in all six components ($\max|\Delta| = 0$) [multi-velocity worker, 2026b].

Figure 20 shows the sampled distributions behind these cases. With a 5% energy spread and $+y$ drift the B_z peak is 2.32050×10^4 T at 16 bins and 2.32049×10^4 T at 64 bins, 0.0003% apart; $B_y = 0$ exactly, which is correct for a purely longitudinal current [multi-velocity worker, 2026b]. With a 5° angular spread the B_z peak moves from 2.48202×10^4 to 2.48213×10^4 T and B_y from 1294.8 to 1295.5 T; the ratio B_y/B_z is 5.22×10^{-2}, and the two bin counts agree to 0.0043% in B_z and 0.056% in B_y [multi-velocity worker, 2026b].

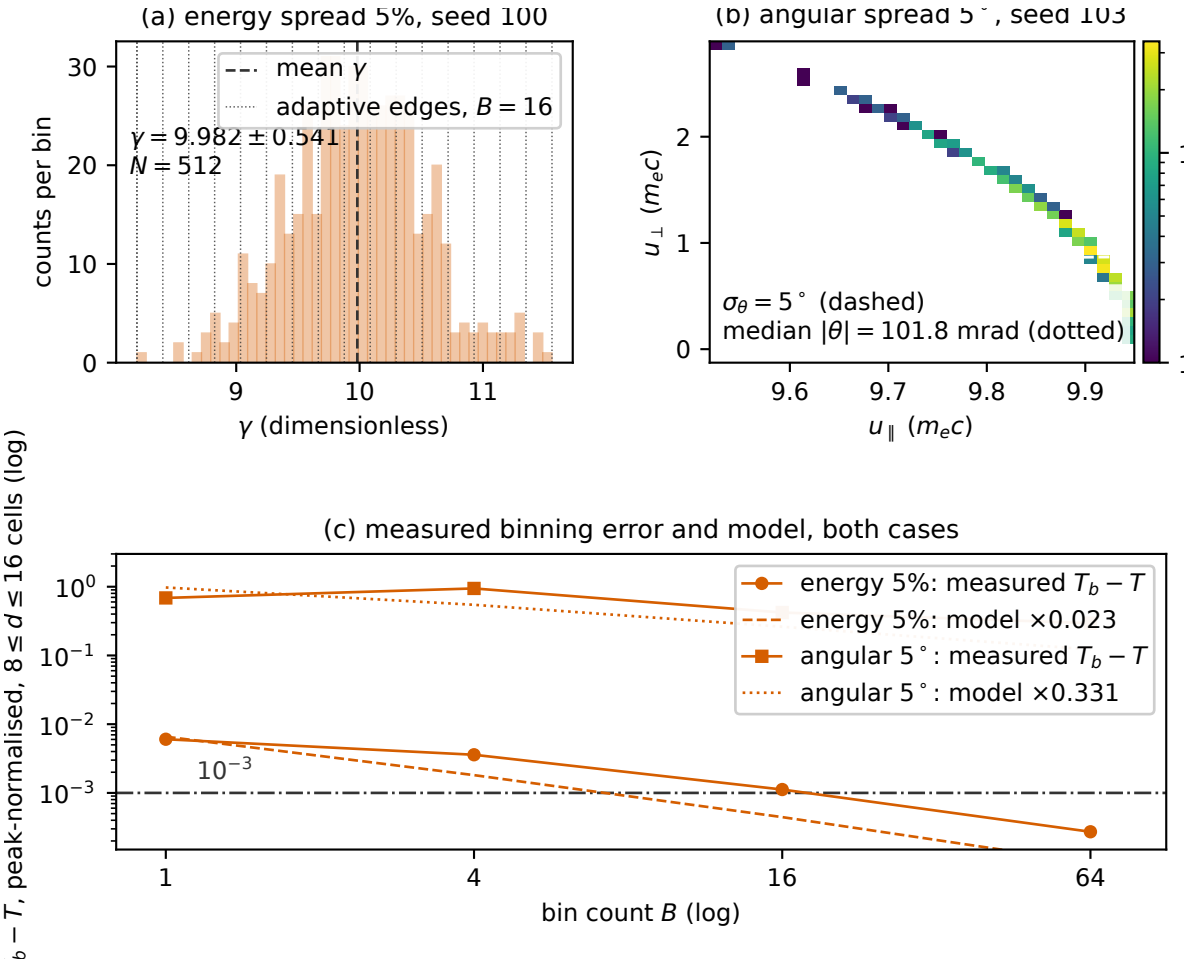


Figure 20: Sampled multi-velocity bunches used by the campaign ($\gamma = 10$, 512 sources, seeds 100/103, generator `02891fe6`) [multi-velocity worker, 2026a]. (a) γ histogram for the 5% energy spread with the $B = 16$ adaptive edges and labelled mean and standard deviation; (b) $u_{\parallel}$ versus $u_{\perp}$ density for the 5° angular spread with the 5° and median-divergence lines; (c) measured per-bin error $T_b - T$ against the model for both spread cases.

1 mrad case against the arbiter. The fluctuation bunch ($N = 200$, $\gamma = 10$, $+y$ drift, 1.046 mrad rms x-divergence, `BDATA2`) was re-run at 64 bins and compared against the order-12 Gauss–Legendre arbiter (unbinned, per-particle u, 698-node slice subset) measured before the fix [multi-velocity worker, 2026c, fluctuation worker, 2026]. On the x-y slice ($z = 32$) picBunch gives $B_y = 7.835$ T and $B_z = 2.3336 \times 10^4$ T, against arbiter values of 7.759 T and 2.344×10^4 T (factors 1.010× and 0.996×); on the z-y slice ($x = 32$) the corresponding values are 7.561 T / 7.519 T and 2.3159×10^4 T / 2.327×10^4 T (1.006× and 0.995×). The 3-D B_y peak is 8.734 T and B_z is 2.3336×10^4 T. The worst component deviation is 1.4% (x-y ratio); every component on every slice is within 1.0%. Before the fix the same 64-bin run gave $B_y = 1.196 \times 10^3$ T, 154× the arbiter, and $B_z = 4.002 \times 10^4$ T, 1.71× the arbiter [multi-velocity worker, 2026c, fluctuation worker, 2026].

FFT buffer defect (fixed in `v0.9.1`). The replicated mesh path allocated its five transform buffers with 3-D `allocUnified` extents. On CUDA and HIP that allocation pads every innermost row to 128 B when the padded M_x is not a multiple of 16 doubles. The per-bin `memset` is pitched: it clears the row data but not the gap bytes, which lie inside the flat prefix the contiguous FFT plan and the flat deposit, kernel, and read-out loops read as data. The velocity-bin loop reuses the buffers, so the second and later non-empty bins read the previous bin's charge from the gaps, and the field grew with the bin count while the deposited charge summed correctly. The scope is exactly that case: more than one non-empty velocity bin on CUDA or HIP with a padded M_x that is not a multiple of 16. A single-velocity run clears a fresh buffer once and is unaffected. `v0.9.1` allocates all five buffers as flat 1-D counts [integration worker, 2026a]. The `picBunchFftSizeCheck` tool (one allocation, a pitched `memset` and a forward/inverse pass per bin, summed against the direct free-space reference) confirms the fix: at the two failing local-box sizes $M = 24$ and $M = 40$ the free-space relative error is 3.5×10^{-16} to 7.0×10^{-16} at every bin count, versus 91.6 and 48.9 with the old 3-D buffers at 128 bins [integration worker, 2026a]. Single-velocity fields are bitwise unchanged between the v0.9.0 and v0.9.1 builds (SHA-256 verified on the 20-cell mesh-only case and the accuracy anchor), so no previously published single-velocity result moves [integration worker, 2026a].

Runtime: $B = 64$, 64^3, 200 particles, `-near-shell fast` ($R = 3$, tile 4, order 5), 248 s on the A30 [multi-velocity worker, 2026c].

Table 15: Whole-field comparison of picBunch against the PIConGPU `referenceOnly` field, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100 [worker, 2026j]. `worst_rel` is peak-normalized; median and p99 are per-point over nodes with $|\text{ref}| > 10^{-3} \times$peak; L2 is pooled.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

7 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Both checks measure agreement against an independently generated reference.

7.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 6), evaluated in PIConGPU in its reference-only field-initializer mode [worker, 2026g]. picBunch is at `339ca7ad` (`experimental/mvp`); `ctest` passes 10/10 [worker, 2026j]. Both sides use identical Yee offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation [worker, 2026j].

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B) [worker, 2026j, Orchestrator, 2026d]. picBunch is about four to six orders of magnitude above that floor; the measured error is concentrated at the core and the beamed peaks, where the TSC support is, while the bulk agrees to below 1–3% and B is exactly zero for $\gamma = 1$ in both [worker, 2026j, Orchestrator, 2026d].

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8 $\times 10^{-3}$ with an L2 of 4.6–5.7 $\times 10^{-2}$; $\gamma = 10$ bulk median 1.4–3.3 $\times 10^{-2}$ but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92 [worker, 2026j]. The forward E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake below the TSC support and the spectral near-field kernel resolution [worker, 2026j].

The sampling groups are distance-from-centroid. The 318-point groups (inside-support, face, outside) were cut by distance from the bunch centroid, in a bunch with a ~ 6-cell spread [Orchestrator, 2026d]. That grouping may understate how close the "outside" points actually are to a source's TSC support, so the per-group numbers are a spatial profile, not a clean inside/outside-of-support split. The per-source dependence in Section 6 uses the explicit support union and is the measurement that separates the per-source floor from any superposition contribution [worker, 2026g].

7.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0, E, B $x/y/z$ at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata [acceptance worker, 2026]. The default precision is uniform `float32` for every field and particle record, so a stock single-precision PIConGPU can load the file [precision worker, 2026]. `v0.9.1` copies every species from the input bunch and carries a single checkpoint-wide `picongpu_idProvider` range [integration worker, 2026b, far-tail worker, 2026, placement worker, 2026]; the `charge` and `mass` records are written rather than omitted, and the species is a single particle patch (Section 3.1). The remaining differences against a genuine PIConGPU checkpoint are non-blocking metadata [acceptance worker, 2026]. For a multi-rank restart the writer splits the species into one patch per output rank: `patch_count` in `[placement]` gives cumulative, disjoint `numParticlesOffset` ranges, with one-cell overlap between neighbouring z regions, and every particle belongs to exactly one patch [mpi worker, 2026a, rank-parallel worker, 2026].

Field arrival is bitwise. A fresh `v0.9.1` checkpoint written from a two-species input (16 electrons and 16 ions, `picongpu_idProvider` startId 0, nextId 32) restarts in PIConGPU with exit 0 under both `patch` modes (`bbox`: offset $(7, 8, 3)$, extent $(8, 10, 6)$; `domain`: offset $(0, 0, 0)$, extent $(24, 24, 12)$), all six field components bitwise-equal, $\max_{\text{abs diff}} = 0.0$, electrons with ids 0 to 15, first-step displacement exactly $\beta c\, dt$ with $\beta = 0.9949874367$ [placement worker, 2026]. A validation-only build with the field solver set to `None` loads the checkpoint and writes step 1; all six components are bitwise-equal, max ULP 0 [acceptance worker, 2026].

A multi-rank restart with the four-patch layout was verified process-level: a 256-particle bunch written as four non-empty patches (cumulative offsets $0, 56, 126, 203$, one-cell overlap) restarts PIConGPU with exit 0, and after one step all six field components are bitwise-equal to the checkpoint, $\max_{\text{abs diff}} = 0.0$ [mpi worker, 2026a].

7.3 PML passthrough and open-boundary restart

On a $24 \times 24 \times 12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the ~ 116 s neuralBunch path to 1.8 s because the checkpoint carries the field [acceptance worker, 2026]. For an open-boundary deck the writer copies the `Convolutional PML` slabs from the input checkpoint verbatim (converting a non-float32 slab to float32), so the PML state arrives with the restart. On a real LWFA checkpoint (638976 electrons, eight `Convolutional PML` groups with six datasets each, $64 \times 128 \times 64$, `-periodic 0 0 1`) the writer copies all eight groups and the restart exits 0; the copied PML datasets are float32 and bitwise equal to the source, initialization is 3.986 s, and the main E and B records are recomputed from the particles and stay float32 [fix-run worker, 2026, Orchestrator, 2026e].

8 Discussion

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method.

What it is good for. In the far field the relative error against the point field is below 10^{-3} on the beam axis from $d = 6$ for every γ [worker, 2026a]. Against the TSC-convolved continuum: $\gamma = 1$, aspect 1 crosses 3×10^{-4} at $d = 12$ (1.1×10^{-4}); $\gamma = 1$, aspect 4 at $d = 32$ (2.6×10^{-4}); $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$ [worker, 2026e,f]. At 2×10^6 particles in a 20-cell box the compute-only total is 64.8 ms, dominated by the deposit (56.5 ms), and flat in γ [scaling worker, 2026a]. Repeated runs are bit-identical (md5 `d3641d64...`); the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14} [worker, 2026e, scaling worker, 2026b].

Measured error in the bunch core. The mesh alone, on the $N = 1000$ slice geometry, has a worst per-component plane error of 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the planes cut the bunch core (Table 10); those core errors are measured against the order-4 driver and so inherit its $\sim 10\times$ overestimate of moving oblique in-support points (Section 6.5). The exact near shell (Section 3.7) is opt-in and its assembly is exact to roundoff; at the measured $R = 3/4$ radii on the $h = 1$ slice grid the residual is the mesh contribution outside the shell, not the tile subtraction (Section 6.7). The arbiter-validated in-support accuracy of the fast backend is 1.6% at the closest moving node, not the 0.31 the driver-based slices reported (Section 6.5, [pr24w53 ref, 2026]). In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M - T$ error at $d = 48$ is 2.6×10^{-4}, the $M - P$ error is 1.5×10^{-2}, and the TSC smoothing $T - P$ is 1.48×10^{-2}; the $M - P$ error is the TSC smoothing, not the deposit (Table 7) [worker, 2026a, Orchestrator, 2026b]. At $r = 8$ cells the node spacing for 10^{-3} perpendicular error at $\gamma = 10$ is $h \approx 1/29$, a 2.4×10^4-fold increase in M^3 FFT work relative to the $h = 1$ run [worker, 2026a, Orchestrator, 2026a]. At aspect 4, $\gamma = 1$, the perpendicular $M - T$ error at $d = 48$ is 2.4×10^{-3} [worker, 2026a]. The cross-validation confirms the same structure: the measured disagreement with the exact reference is per-source, local to the TSC support, and independent of particle count [worker, 2026g, Orchestrator, 2026d].

Performance. The cost is two regimes with a crossover set by the box, not the particle count [scaling worker, 2026a]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows. At large boxes the FFT and kernel dominate: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× [scaling worker, 2026a]. The compute timings are single-GPU on one `rosi_v100` host; the same mesh is 0.052 s on one H100 (Section 4) [release worker, 2026]. The rank scaling is measured on an idle V100 window: the component-parallel mesh plateaus at 1.20× (replicated deposit and source FFT), the tile-parallel near shell at 1.34× (static tile split plus all-reduce), and the collective writer improves to 1.71× at two ranks and falls to 0.48× at four, where it is I/O-bound (Section 4.1) [mpi worker, 2026a, rank-parallel worker, 2026]. The near-field cost is a separate wall, measured end-to-end through the CLI on one H100: 281.0 s for the support-only shell ($R = 1.5$) and 1510.7 s for the four-cell shell ($R = 4$) at 2×10^6, against a mesh-only wall of 2.48 s (Section 6.8) [release worker, 2026]. The shell radius is the dominant cost lever, and the device backend runs the same fast algebra on the CUDA device, agreeing with the host fast path to 5.4×10^{-16}–8.0×10^{-16} (Section 6.9); the distributed FFT makes the output field slab-local at the cost of `MPI_Alltoallv` transposes that are 74–85% of the FFT-plus-transpose time, a memory feature rather than a speed feature (Section 6.10).

Measured error by configuration. In floored relative per-component error against the TSC-convolved continuum, the measured values at $d = 12$ are 1.1×10^{-4} for $\gamma = 1$ aspect 1 and 2.6×10^{-4} for $\gamma = 1$ aspect 4; the axial component is below 3×10^{-4} from $d = 6$ for every γ [worker, 2026e,a, Orchestrator, 2026a]. The perpendicular direction at $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$, the measured values at those distances [worker, 2026e,a,f]. The fast near backend, against the same continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend (Section 6.6) [fast-near worker, 2026, release worker, 2026]. Each statement in Section 6 names which metric (floored, unfloored, absolute) supports it; the floored column is electric-only at $\gamma = 10$ because the floor excludes the magnetic field, and the judge metric $|\text{ref} - \text{mesh}|/(1 + \text{max})$ reports $d = 2$ for every case and does not discriminate. The radii in Table 9 are axis directions only; oblique directions are $\sim 1.5\times$ larger. At a fixed distance in cells the controlling length is the beaming cone width d/γ; for $\gamma \geq 50$ the cone is sub-cell and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls to about 1 (Section 6.13) [worker, 2026c].

Structural assumptions. The method assumes a single shared velocity (or a set of binned velocity classes, one convolution per class, Section 6.16); the near shell needs no binning because each pair carries its own velocity. The computed quantity is the TSC-convolved uniformly-moving point-charge field; the convolution is free-space (no boundary condition other than zero fill beyond the far-tail switch); and there is no radiation term, because the motion is uniform. A bunch with a velocity spread requires one convolution per velocity class; the measured binned behaviour (v0.9.1, bin-independent to 0.004% for the 5% energy spread and 1.4% against the arbiter for the 1 mrad case) is in Section 6.16, and the validated shipped scope is a single shared $\mathbf{u}$ (Section 2).

9 Conclusion

`picBunch v0.9.1` computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, driven by a single TOML configuration (Section 3.1), and writes a uniform float32 checkpoint that restarts under a stock PIConGPU, periodic and open-boundary, single- and multi-patch. The release passes 19/19 ctest with zero compiler warnings on a V100 `sm_70`, an H100 `sm_90`, and a gfx1100 HIP build [release worker, 2026].

The opt-in near shell assembles exactly to roundoff, and its tile subtraction is not the limiting term; the seam excess at the fast/exact crossing is 2.03×10^{-10} per component for a stationary single particle (2.8×10^{-10} for the 64-source cloud) [fast-near worker, 2026, release worker, 2026]. The fast backend, an eight-term `asinh` box plus a plain Gauss–Legendre remainder, meets 10^{-3} stationary (7.99×10^{-4} interior) and moving axial (1.69×10^{-4} interior); the moving oblique interior is 8.15×10^{-2}, the measured limit of the fast backend [fast-near worker, 2026, release worker, 2026]. At 2×10^6 particles the mesh-only CLI wall is 2.48 s on one H100 (compute 0.116 s), deterministic (md5 `d3641d64...`); the near shell is 281.0 s end-to-end support-only at 7.99×10^{-4} and 1510.7 s end-to-end four-cell at the same interior error, the shell radius the dominant cost lever (Section 6.8). The device backend runs the fast algebra on the CUDA device, agreeing with the host fast path to 5.4×10^{-16}–8.0×10^{-16} (Section 6.9), and the distributed FFT makes the output field slab-local with the transposes dominating the FFT-plus-transpose time, a memory feature (Section 6.10).

In the far field the mesh is below 3×10^{-4} from $d = 12$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1, the measured values are 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$ [worker, 2026e,f]. The γ-landscape measurement shows the error growing steeply with γ at a fixed distance in cells, with the beaming cone width d/γ the controlling scale: at $d = 8$ cells, $h = 1$, the error is 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises (Section 6.13) [worker, 2026c]. The MPI compute is component-parallel mesh and tile-parallel near shell with one all-reduce each, bitwise identical at 1/2/4 ranks, with the replicated deposit and source FFT capping the mesh speedup and the collective writer I/O-bound at four ranks (Section 4.1) [mpi worker, 2026a, rank-parallel worker, 2026]. The method is a far-field tool: cheap, deterministic (md5 `d3641d64...`), and portable across the three backends; the measured in-support error at $\gamma \geq 50$ is reported in Section 6.13.

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INT2 placement worker. picbunch v0.5.0 release: placement interface (int2), 2026. Run pr24-20261102. INT2/RESULTS.md. v0.5.0, commit 7b4f1766, tag 3e083fef. TOML interface, charge-weighted centring, float32 default, charge/mass records, patch = bbox or domain. Restart: bbox and domain exit 0, six components bitwise, ids 0-15, idProvider 0/32, first step beta*c*dt. ctest 18/18.

INT3 mpi worker. picbunch v0.6.0: rank-parallel compute and release (int3), 2026a. Run pr24-20261102. INT3/RESULTS.md. v0.6.0, commit e4c7325d. Component-parallel mesh and tile-parallel near shell with all-reduce; bitwise 1/2/4-rank equality. Clean rosi_v100 scaling: near shell 1.24x/1.34x, mesh 1.20x/1.20x, write 1.71x/0.48x at 2/4 ranks. Multi-patch restart: 256 particles, four patches, bitwise. Release tag df7de342.

IMPL3 precision worker. Uniform checkpoint precision and stock-picongpu restart (impl3), 2026. Run pr24-20261102. IMPL3/RESULTS.md. precision = float32 default for every field and particle record; float64 uniform opt-in with a picBunchPrecisionWarning attribute. float32 restart on rosi_v100: exit 0, six components bitwise, ids 0-15, first step beta*c*dt. Branch experimental/placement, commit b68a141bb.

SB1 scaling worker. Measured scaling of the picbunch field pipeline (sb1), 2026a. Run pr24-20261102. SB1/RESULTS.md. V100 timing matrix, Coulomb limit, distance sweep. Commit dc69c16a.

N2b worker. picbunch mesh vs the tsc-convolved continuum and the point field (n2b), 2026a. Run pr24-20261102. N2b/RESULTS.md. Reference-separated accuracy, scale invariance, crossover table. Arbiter: indfield.py.

N1b worker. picbunch mesh convolution measurements, cpu (n1b), 2026b. Run pr24-20261102. N1b/RESULTS.md. Wraparound control, R2C/C2R vs C2C, padding vs distance, kernel sampling rule, CPU cost model. CPU only, no GPU.

ER1b worker. Single-particle tsc field error landscape, gamma 10 to 100 (er1b), 2026c. Run pr24-20261102. ER1b/RESULTS.md. 96-cell box, cell 1e-7 m, aspect 1, axial and oblique 45 deg, h=1 and h=1/4, jureca_h100. absn at d=8, h=1: 4.4e-2 (g10), 1.05 (g50), 5.47 (g100); h=1/4: 6.9e-5, 5.6e-2, 0.82. Slope p 1.85 (g10) to 1.0 (g>=50). Beaming cone width d/gamma sub-cell at g>=50; 1e-2 needs h<=0.002 at g50, h<=0.001 at g100; shell radius O(gamma) cells.

NS2 near-shell worker. picbunch exact per-pair near shell (ns2), 2026. Run pr24-20261102. NS2/RESULTS.md. F = F_mesh(all) - F_mesh(near) + F_exact(near), opt-in –near-shell exact, per-tile local FFT, vendored FP64 TSC reference. Envelope and cost (3.1e-4 s per pair) in the text. Commit 9733d232.

FIX1 far-tail worker. picbunch v0.4.1 release: near-shell cli and far-tail margin (fix1), 2026. Run pr24-20261102. FIX1/RESULTS.md, margin_table.txt. v0.4.1, commit 02d2d06e, tag 9e4fd7dc. Near-shell CLI parsed; quadrupole far-tail bound and 6R switch (switch error, 512-cubed MPI equality in the text). Checkpoint: all species, checkpoint-wide idProvider, restart bitwise. ctest 16/16 (rosi sm_70, jureca-h100 sm_90, hal gfx1100 HIP).

CFPORT fast-near worker. Fast closed-form near-field evaluator in picbunch (cfport), 2026. Run pr24-20261102. CFPORT/RESULTS.md. –near-shell fast: eight-term asinh box (O(1)) plus plain Gauss-Legendre on the subtracted integrand (S-W*)K over the TSC knot panels, order 5 (8000 nodes) in-support; moving axial gamma-stretched box; moving oblique rest-frame octant order 8 (4096 nodes). Accuracy vs indfield: stationary interior 7.99e-4, face 1.96e-5, outside 1.57e-7; moving axial interior 1.69e-4 (order 8); moving oblique interior 8.15e-2 (order 5, the limit). Seam 2.03e-10; mesh-only bitwise identical between fast and exact. Branch experimental/fast-near 9a7e574a, from v0.5.0.

pr24w53 ref. indfield.py: independent tsc-convolved continuum reference, 2026. sha256 a1355980. Tensor Gauss-Legendre with TSC-knot split and a singularity-cancelling power map; the independent truth for the mesh.

CF1 closed-form worker. Closed-form in-support field and pair counts (cf1), 2026. Run pr24-20261102. CF1/RESULTS.md. Eight-term asinh box re-verified; 2e6 pair counts: support 1.7e8, 3-cell 1.7e9, 4-cell 3.4e9 (85/850/1700 neighbours per source); H100 rate 1.837e7 pairs/s order 5, 3.983e6 order 8 axial.

K3 evaluator worker. In-support evaluator port and per-pair cost ladder (k3), 2026. Run pr24-20261102. K3/RESULTS.md. O(1) asinh box verified against a converged trapezoid reference (-2.6333 vs -2.6348); octant-split remainder; power-transform quadrature the validated moving path (1.000000/1.000000/1.000001/1.000040 at gamma 1/2/10/100, 1.5e-6 at box order 32); no validated closed form for the moving case. Per-pair costs, rosi_v100 single thread: ported box8/rem8 5.4e-5 s; accurate moving 3.1e-4 s out of / 6.9e-3 s in support; upstream adaptive (1.07e9 nodes) 1.62 s.

NS4b worker. Near-shell assembly diagnosis: subtraction is exact (ns4b), 2026d. Run pr24-20261102. NS4b/RESULTS.md. Tile-centre/boundary residuals 2e-12 to 1.5e-11 SI (4e-16 rel); local vs global FFT 1e-15; 100-particle reproduction: assembled 4.6e-13 vs plain mesh 5.6e-1. The 1.36e-1 plane error is mesh-outside-the-shell, not assembly. v0.4.1 (02d2d06), rosi_v100.

INT4 release worker. picbunch v0.7.0: fast near shell and release (int4), 2026. Run pr24-20261102. INT4/RESULTS.md. v0.7.0, commit a152c518, tag 25d6ca8b. Merges fast-near 9a7e574a and hip-teardown 833b6275. 18/18 ctest, zero warnings, on rosi_v100 sm_70, jureca_h100 sm_90, hal gfx1100 HIP. H100 2e6/20-cell: mesh 0.052 s; near 1.8466e7 pairs/s stationary o5, 3.9977e6 moving axial o8; support-only 9.26 s at 7.99e-4, 3-cell 92.1 s, 4-cell 184.2 s, moving axial 42.6 s at 1.69e-4. Seam 2.03e-10; mesh-only md5 identical between backends.

B1 reference-field worker. Pr #24 referenceonly cost and a working reference-field path (b1), 2026. Run pr24-20261102. B1/RESULTS.md. Bunch generator generate_bunch.py –model gamma –seed 42 –grid 64 64 32 –cell-size 1e-7 –centroid 32 32 16 –sigma 6 6 6 –weights uniform; variants gamma{1,10}-1k (N=1000) and -100 (N=100); readback bunches_verification.json (weight sum 10000, mean/min/max gamma 1.0 and 10.0); gamma{1,10}_1k_reference .h5/json: full field N=1000, 32x32x16, cell 2e-7 m, tol 1e-4. rosi_v100 (V100, CUDA 12.6).

LARGE1 large-domain worker. Large-domain slice figures, 128x128x64 (large1), 2026. Run pr24-20261102. LARGE1/RESULTS.md. 128x128x64, cell 2e-7 m, N=100, jureca_h100, reference tol 1e-3 (max error 1e-3). Edge field at most 0.103 of peak (gamma10 z-y Ez); core error gamma10 9.27, gamma1 0.759 (method error).

MV2 multi-velocity worker. Binned multi-velocity binning: measured binning error and model calibration (mv2), 2026a. Run pr24-20261102. MV2/RESULTS.md. Branch experimental/multivel at 02891fe6bd9c34eea5479e91ecebbd54a44a47f7 (parent 2c736df), jureca_h100 jrc0880, H100 PCIe, CUDA 13.4, sm_90. 48^3 grid, 512 sources, seed 100 (energy) and 103 (angular). Four defect fixes (dangling pointer, parse_bands, model floor, unit-test build); comparison-artefact diagnosis; single-velocity bitwise sanity; Tb-T binning error and total error with fast shell R=3 vs bin count; model calibration C=1.09 (energy, predicts to 10–55%) vs C=124 non-monotonic (angular); runtime mesh-only 5.26/5.62 s at 1e6/2e6 in 48^3 with 64 bins, 184 s for the 4-cell fast shell at 2e6 in a 20-cell bunch. ctest 19/19.

MVRERUN2 multi-velocity worker. Minimal multi-velocity bin-independence re-measurement on v0.9.1 (mvrerun2), 2026b. Run pr24-20261102. MVRERUN2/RESULTS.md. picBunch v0.9.1 (commit c0bf82c2251448abfaad66aa03b18ab3a9ab9322), tag v0.9.1 (262927baf42f26bd3fae21ac86732c3f84960bbd). hal (c692209a108c, NVIDIA A30 24 GB, sm_80, CUDA 13.4.92). 64^3 grid, cell 1e-7 m, 200 sources, BDATA2 bunches (sigma=6 cells, centroid (32,32,32), gamma=10, +y drift), –near-shell fast R=3 tile 4 order 5, –deterministic-deposit. Cases: mono (zero spread), e5 (5e5 Bz 2.32050e4 / 2.32049e4 (16 / 64 bins), By = 0 exactly; a5 Bz 2.48202e4 / 2.48213e4, By 1.2948e3 / 1.2955e3; ratio By/Bz 5.22e-2, bin-independent to 0.0043components, max|diff|=0. Wall: mono 195.8 s, e5 224.7 / 250.1 s, a5 229.2 / 250.0 s.

MVRERUN3 multi-velocity worker. 1 mrad case: v0.9.1 64-bin field vs the pre-fix order-12 arbiter (mvrerun3), 2026c. Run pr24-20261102. MVRERUN3/RESULTS.md. picBunch v0.9.1 (c0bf82c), hal A30 sm_80. Bunch: fluc1mrad.bunch.h5 (BDATA2 fluctuation, N=200, -100 pC, gamma=10, +y drift, centroid (32,32,32) cells, sigma=6 cells, x-only 1.046 mrad rms divergence). 64^3 grid, cell 1e-7 m, 64 bins (64x1x1), –near-shell fast R=3 tile 4 order 5. 3-D peaks: By = 8.734 T, Bz = 2.33363e4 T, ratio 3.74e-4. Slice peaks (xy z=32): By = 7.8346 T, Bz = 2.33363e4 T, ratio 3.357e-4; (zy x=32): By = 7.5615 T, Bz = 2.31587e4 T, ratio 3.265e-4. Arbiter (FLUC1, order-12 Gauss-Legendre, unbinned, 698-node subset): xy By = 7.759 T, Bz = 2.344e4 T, ratio 3.31e-4; zy By = 7.519 T, Bz = 2.327e4 T, ratio 3.23e-4. Worst deviation 1.421.0arbiter). Compute wall 247.98 s.

VAR1 divergence worker. Real bunch: forward divergence makes every field component nonzero (var1), 2026. Run pr24-20261102. VAR1/RESULTS.md. Base case = mid-bunch g10 (gamma10-1k, N=1000, seed 42, sigma 0.6 um/axis, electron charge, gamma = 10); field grid 32x32x16 nodes, cell 2e-7 m, origin (0,0,0), CUDA, –near-shell none; paraxial +y model u = u0*(tx,1,tz)/|(tx,1,tz)|, tx,tz N(0, sigma), seeds 43, sigma = 1 mrad and 1 deg; per-macro charge -1.602176634e-19 C, centroid (3.197, 3.208, 1.615) um. rosi_v100 (gv025, V100). The VAR1b N=200 subset (first 200 rows of the 1 mrad bunch) never ran.

MV1b multi-velocity worker. Multi-velocity binning: comparison artefact and calibrated error (mv1b), 2026d. Run pr24-20261102. MV1b/RESULTS.md. Measurement driver picBunchVelocityBinsMeasure, 48x48x48 grid, cell 1e-6 m, 512 sources, 3-cell position cloud; cases g10_e5 (sigma_g/g=0.05, seed 100), g50_e5 (seed 101), g10_a1 (seed 102), g10_a5 (5 deg angular spread, seed 103); comparison artefact diagnosis and per-box model sums.

M1 mpi worker. picbunch global-domain output and mpi (m1), 2026b. Run pr24-20261102. M1/RESULTS.md. –global-grid/–global-origin, chunked writer, PICBUNCH_ENABLE_MPI. 512 cubed bitwise equality across rank counts; rosi_v100, OpenMPI 4.1.6, openPMD 0.17.1.

MPI2 rank-parallel worker. Rank-parallel compute and multi-rank patches (mpi2), 2026. Run pr24-20261102. MPI2/RESULTS.md. Component-parallel mesh (replicated deposit and rho FFT) and tile-parallel near shell (one all-reduce each); bitwise 1/2/4-rank equality. Multi-rank patches: patch_count=4, cumulative disjoint numParticlesOffset. jureca_h100 sm_90, branch experimental/mpi-compute 79fa8f26.

SB1 scaling worker. Coulomb-limit control (sb1), 2026b. SB1/RESULTS.md and PF/data/coulomb.csv. u = 0, one particle at a node, far nodes at d = 8, 16, 24, 32, six axis directions. Worst relative error 4.88e-14 (max absolute 2.28e-25); threshold 1e-12: PASS.

W4b worker. Independent validation of the shipped picbunch (w4/w4b), 2026e. Run pr24-20261102. W4b/RESULTS.md. picBunch at bfaa1e26 (experimental/mvp), rosi_v100, ctest 8/8. Shipped CLI vs indfield.py order 12 refine 3, relative per-component error per d = 2 to 32 cells, gamma 1 and 10, aspect 1 and 4.

Orchestrator. Decision d11: v0.1 accuracy scope, 2026a. Run pr24-20261102. DECISIONS.md, D11. Not defensible: gamma <= 10 meets 1e-3. Defensible: gamma <= 1 aspect 1; gamma <= 10 axial only; gamma 100 axial only.

Orchestrator. Decision d12: Sb1's gamma-10 floor was a reference artefact, 2026b. Run pr24-20261102. DECISIONS.md, D12. T-P = 1.48e-2; M-T = 2.6e-4. All accuracy statements must name the reference.

Orchestrator. Decision d8: gamma-100 outside-band 9.43e-2 is unresolved, 2026c. Run pr24-20261102. DECISIONS.md, D8. Pre-existing 9.43e-2 at 1.6 to 6 cells, gamma 100 axial, outside the near box.

V2b worker. No-floor accuracy audit of the shipped picbunch executable (v2b), 2026f. Run pr24-20261102. V2b/RESULTS.md. Three-number metric (floored, unfloored, absolute) per case and distance; gamma10 B excluded by the floor (B 1e-10 of the local field), unfloored B 9.683e-4 at d=32. picBunch main v0.3.0 a12ce3fa.

DIR1 beam-direction worker. Mid-bunch 2-d slice figures with the beam along +y (dir1), 2026. Run pr24-20261102. DIR1/RESULTS.md, data/bunch_y.npz, data/pic_y.h5. picBunch v0.7.0, N=1000, 32x32x16, cell 2e-7 m, mean u=(0,9.94987437,0), gamma=10; independent FP64 exact TSC reference, order 4, max level 5, tol 1e-3. Planes x-y at z=1.60 um, z-y at x=3.20 um. B_y is identically zero.

SL3 slice worker. Mid-bunch slices and near-shell acceptance (sl3), 2026a. Run pr24-20261102. SL3/RESULTS.md. N=1000, 32x32x16, cell 2e-7 m, rosi_v100, v0.4.0 (bc0037c5). Default-mesh and near-shell (R=3/4, mesh nodes, h=1) plane errors, Gauss-law residual, runtime; the 3/4 envelope assumed h=1/4, so at h=1 the shell is 4x too small. Driver HEAD e79ddb389.

SL2 slice worker. Mid-bunch 2-d slice figures, stationary gamma 1 (sl2), 2026b. Run pr24-20261102. SL2/RESULTS.md, data/slices_g1.npz. Same geometry and reference as SL1 (driver max error 9.075e-4, 14 not converged). B identically zero; electric components only.

DIST1 distributed FFT worker. Distributed fft: z-slab decomposition with alltoallv transposes (dist1), 2026. Run pr24-20261102. experimental/distributed-fft at d602484d. z-slab decomposition, one slab per rank, MPI_Alltoallv transposes around each 2-D transform. Agreement with the replicated path at 2.2×10^{-16}, written outputs bitwise equal at 1, 2, 4 ranks. On a 640-cubed output, host memory per rank is 41.1 GB at 2 ranks and 20.6 GB at 4. Transposes are 74-85% of the FFT-plus-transpose time. Output field slab-local, the bunch replicated.

SCHEM1 schematic worker. Algorithm-region schematic of the field evaluation (schem1), 2026. Run pr24-20261102. SCHEM1/figs, make_schematic.py. Two panels: TSC support / near shell / mesh regions with the tolerance-driven shell radii (3e-4 crossover, 11.8 nodes gamma1, 8.8 gamma10, h=1, aspect 1) and the measured per-component error vs distance (dsweep.csv, SB1).

W7t worker. Reference-quantity identification and particle-count scaling (w7t), 2026g. Run pr24-20261102. W7t/RESULTS.md. The reference is the TSC-convolved field, the same quantity picBunch computes. N-count scaling: disagreement is per-source, not superposition; B identically zero for the stationary bunch.

A1b worker. picbunch anisotropic-cell correctness, per-axis sampling (a1b), 2026h. Run pr24-20261102. A1b/RESULTS.md. Per-axis sampling is correct (FFT equals the direct discrete TSC sum to 4e-16 on cell 1,1,4); the anisotropic gap is the TSC convolution itself (continuum 0.9638 times point Coulomb, 3.2 percent; mesh matches continuum to 0.44 percent at 12 cells). picBunch 339ca7ad.

A1c worker. picbunch anisotropic-cell field correctness, cli (a1c), 2026i. Run pr24-20261102. A1c/RESULTS.md. v0.1.3 da3cd42. The stretched axis (dz=4 m) of a (1,1,4) grid matches indfield.py to about 1e-3, the cubic-control scale; the short x/y axes sit in the near-field TSC regime. Per-geometry quantification, not an anisotropy defect.

INT7 integration worker. Replicated fft buffer fix integration and v0.9.1 release (int7), 2026a. Run pr24-20261102. INT7/RESULTS.md. Repository psychocoderHPC-bot/picBunch. Fixed commit 6c81ecae, merge 9963881178, follow-up 8270bb8, release c0bf82c2251448abfaad66aa03b18ab3a9ab9322, tag v0.9.1 (annotated 262927baf42f26bd3fae21ac86732c3f84960bbd). Defect: 3-D allocUnified pads innermost row to 128 B when padded Mx is not a multiple of 16 doubles; pitched per-bin memset leaves gap bytes uncleared; contiguous FFT plan reads them as data; buffer reuse across velocity bins feeds bin 0 charge into later bins; field grows with bin count. Fix: allocate all five replicated-path transform buffers as flat 1-D std::size_t counts. Scope: multi-bin CUDA/HIP only; single-velocity unaffected. picBunchFftSizeCheck validation: M=24 and M=40 free-space relative error 3.5e-16 to 7.0e-16 at all bin counts (was 91.6 and 48.9 at 128 bins with 3-D buffers). Single-velocity regression: 20-cell mesh-only and accuracy anchor bitwise unchanged (SHA-256 verified). 19/19 CTest on hal (HIP gfx1100) and fwk394 (CUDA 12.5, sm_70).

FLUC1 fluctuation worker. Fluctuation-case multi-velocity binning, 32/64/128 bins (fluc1), 2026. Run pr24-20261102. FLUC1/RESULTS.md. Pre-fix v0.8.0. BDATA2 fluctuation bunch (N=200, -100 pC, gamma=10, +y drift, 1.046 mrad rms x-divergence), 64^3 grid, cell 1e-7 m. 32/64/128 velocity bins (32x1x1, 64x1x1, 128x1x1). 64-bin 3-D peaks: By = 1.196e3 T, Bz = 4.002e4 T (growing with bin count: Bz 2.344e4 at 32 bins to 5.581e4 T at 128 bins). Order-12 arbiter (unbinned, per-particle u, 698-node slice subset): xy By = 7.759 T, Bz = 2.3436e4 T; zy By = 7.519 T, Bz = 2.327e4 T. The 64-bin binned field was 154x the arbiter in By and 1.71x in Bz due to the replicated FFT buffer defect fixed in v0.9.1.

W7r worker. picbunch cross-validation against the picongpu field reference (w7r), 2026j. Run pr24-20261102. W7r/RESULTS.md. picBunch @ 339ca7ad. 318 staggered points (64x64x32, cell 1e-7 m) and whole-field (N=1000, 32x32x16, cell 2e-7 m); groups distance-from-centroid (D13).

Orchestrator. Decision d13: cross-validation does not show agreement at the reference floor, 2026d. Run pr24-20261102. DECISIONS.md, D13. 3.66e-2 (gamma 1), 5.16e-1 (gamma 10 E/y) vs floor 1.19e-6 (E), 1.43e-6 (B); no agreement claim with PIConGPU.

W7B acceptance worker. picbunch checkpoint acceptance on picongpu pr 24 (w7b), 2026. Run pr24-20261102. W7B/RESULTS.md. Early acceptance state: 16-particle periodic deck restarts exit 0, six components bitwise (0 ulp) via a solver=None load; init 1.8 s vs 116 s. Fix commit 0c9427dee82e01c832c42a8358033b701386f37e.

INT1 integration worker. picbunch v0.4.0 integration release (int1), 2026b. Run pr24-20261102. INT1/RESULTS.md. Integration commit bc0037c5, tag v0.4.0; checkpoint writer copies every species, MPI 512-cubed bitwise, two-species restart bitwise.

W7 fix-run worker. Lwfa open-boundary restart with pml passthrough (w7fix), 2026. Run pr24-20261102. Real LWFA checkpoint, 638976 electrons, eight Convolutional PML groups (48 datasets), 64x128x64, periodic 0 0 1, rosi_v100. picBunch copies all eight PML groups bitwise (float32); restart exit 0, initialization 3.986 s. Branch experimental/mvp bed4344c, W3f tree.

Orchestrator. Decision d14: checkpoint acceptance is proven, including pml and scalar records, 2026e. Run pr24-20261102. DECISIONS.md, D14. 638,976 particles; 48 PML bitwise; restart exit 0; initialization 3.986 s; periodic case 0 ulp. Limitation: PML copied from the same input file.